

USTHB University
Faculty of Computer Science
AI&DS Department

DevOps & Cloud Computing

02- Cloud Computing Architecture





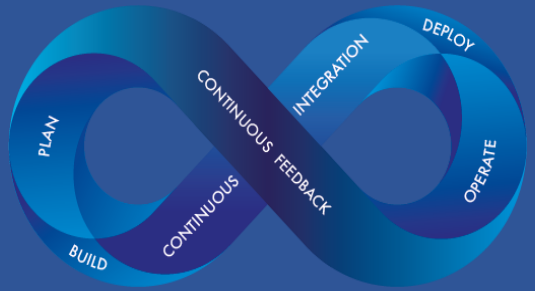


Engineers (5th Year) | AMEUR IMANE | 2026/2027



COURSE OUTLINE

What you'll learn in this course



06 **Cloud architectures**
Reference architecture, layers, tiers

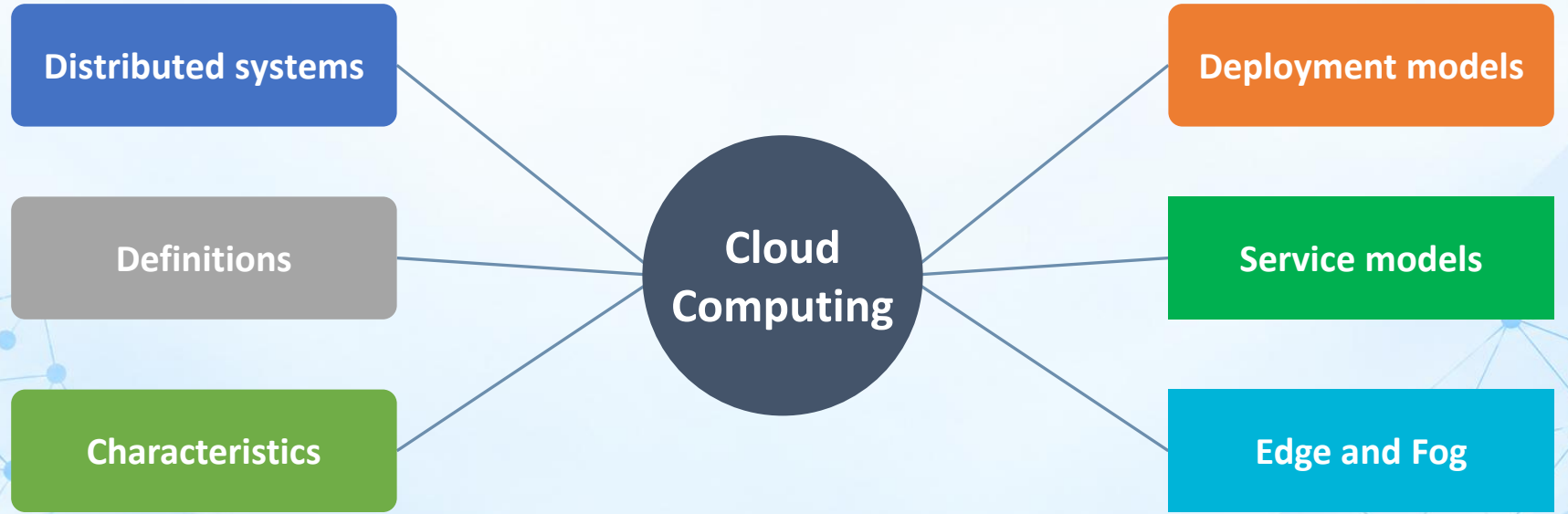
07 **Enterprises facing the cloud**
Adoption, drivers, challenges, migration

08 **Providers and clients**
Roles, major providers, SLAs

09 **Multi-cloud environment**
Multi-cloud vs hybrid

10 **Virtualization and cloud**
Technical foundation of the cloud

Overview: Chapter Mind Map



Cloud Computing Evolution



Mainframes



PC



Client/ Server



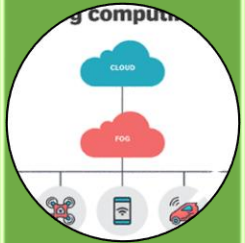
Web Computing (SaaS)



Cloud Computing



Edge Computing



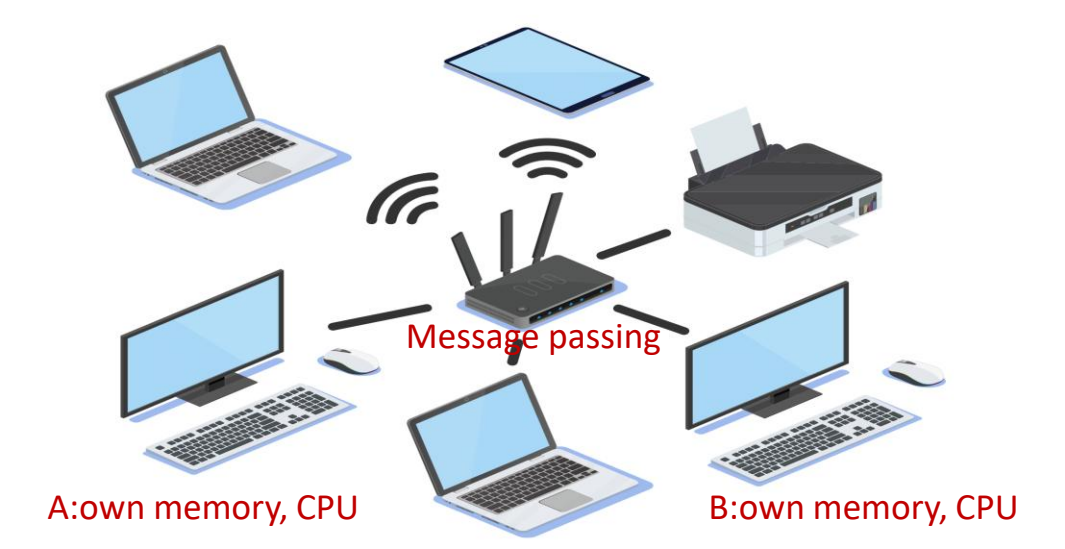
Fog computing

1. Recap of Distributed Systems



Concepts, architectures, object/component/service, SOA

What is a Distributed System?



A collection of independent computers that appears to its users as a single coherent system
Nodes communicate and coordinate only by exchanging messages
The cloud is, at its core, a very large distributed system

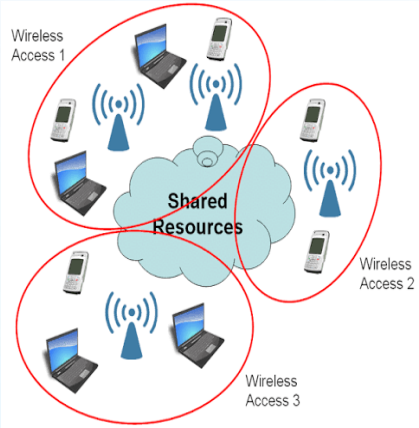
Why build Distributed Systems?



- **Scalability**
  Add more machines to handle more load



- **Availability**
  No single point of failure



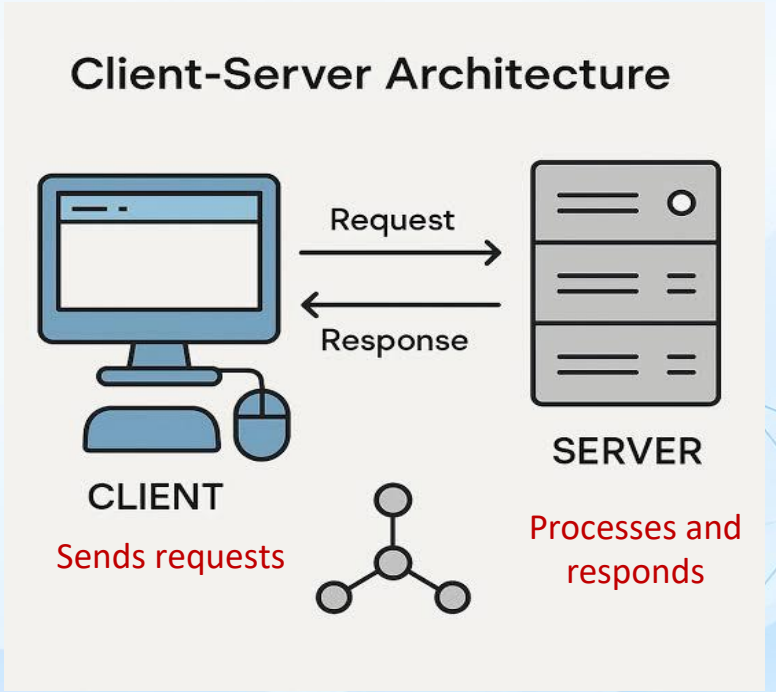
- **Resource sharing**
  Combine the power of many machines



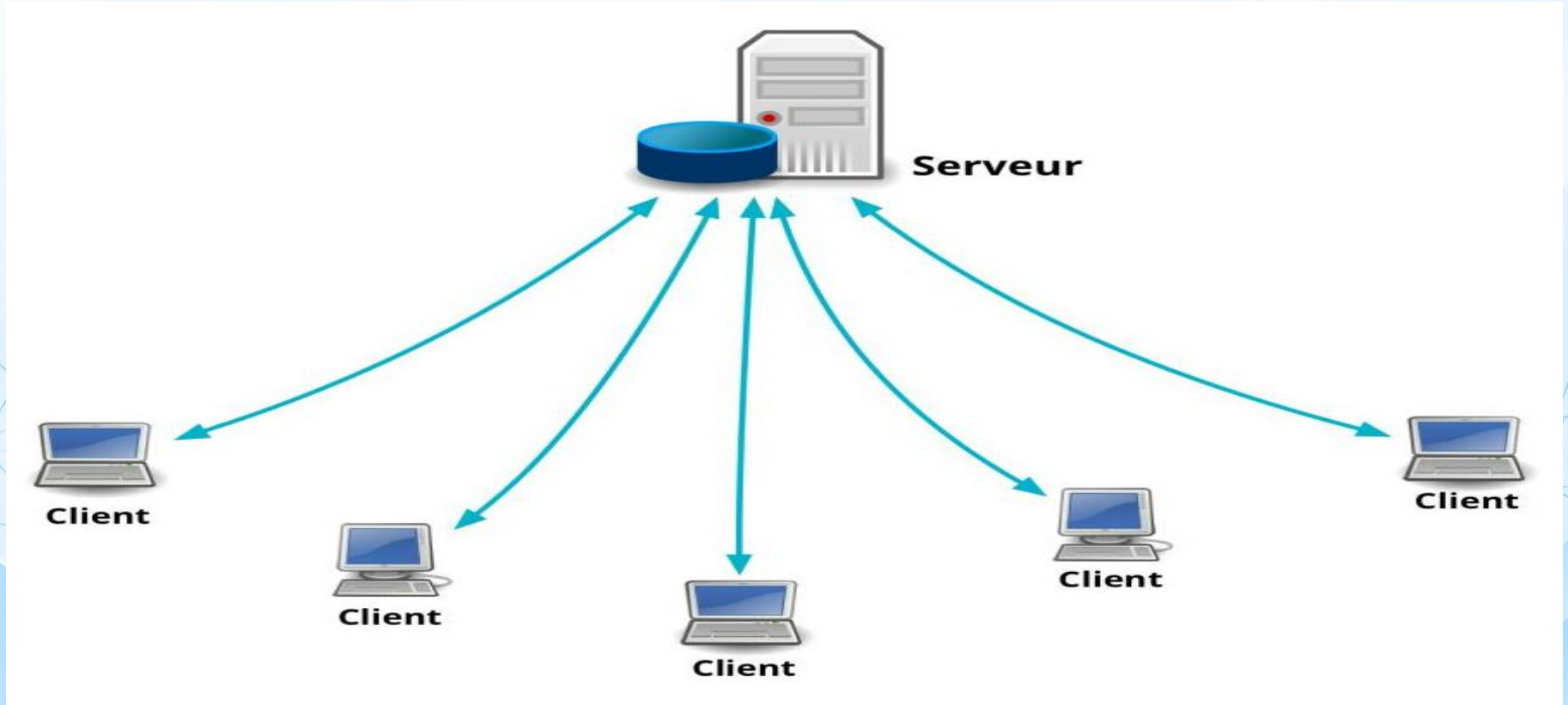
- **Geographic reach**
  Serve users close to where they are

Client-Server Architecture

- Centralized model: the server holds data and logic, clients consume the service,
- Simple to reason about, but the server can become a bottleneck or single point of failure,
- Most classic cloud services are still client-server at heart,

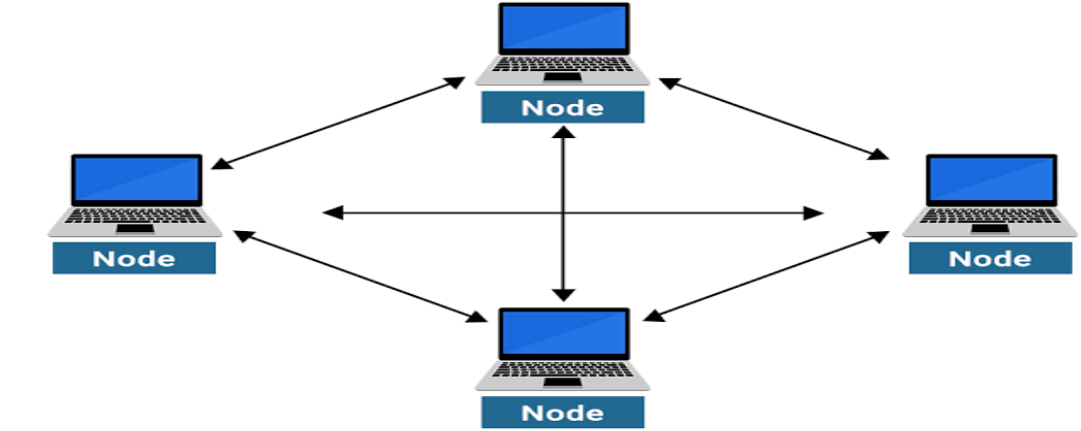


Client-Server Architecture



Peer-to-Peer (P2P) Architecture

Peer-to-peer (P2P) is a distributed architecture in which computers, called peers, communicate and share resources directly with one another, without depending on a single central server for every operation.

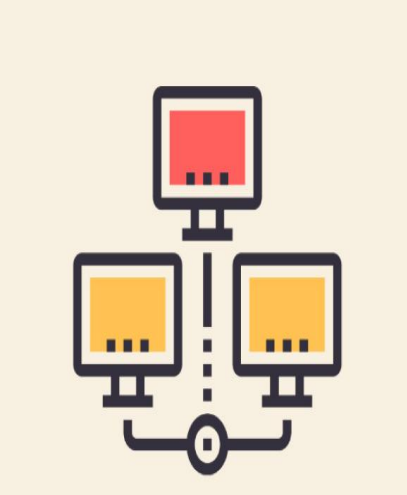


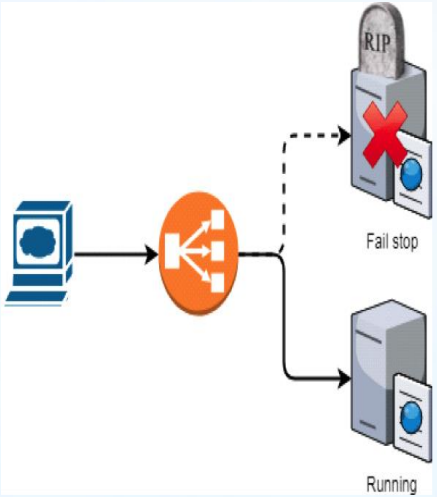
P2P vs Client-Server

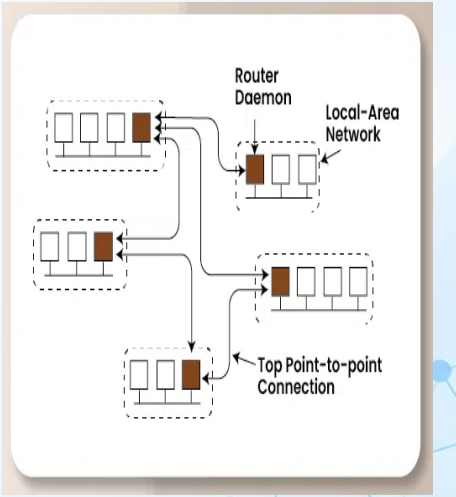
Criterion	Client-server	Peer-to-peer
Roles	Fixed: client or server	Every node can be both
Single point of failure	Yes (the server)	No, more resilient
Complexity	Simpler to manage	Harder to coordinate
Example	Web hosting, cloud APIs	File sharing, blockchain

Core Challenges of Distributed Systems



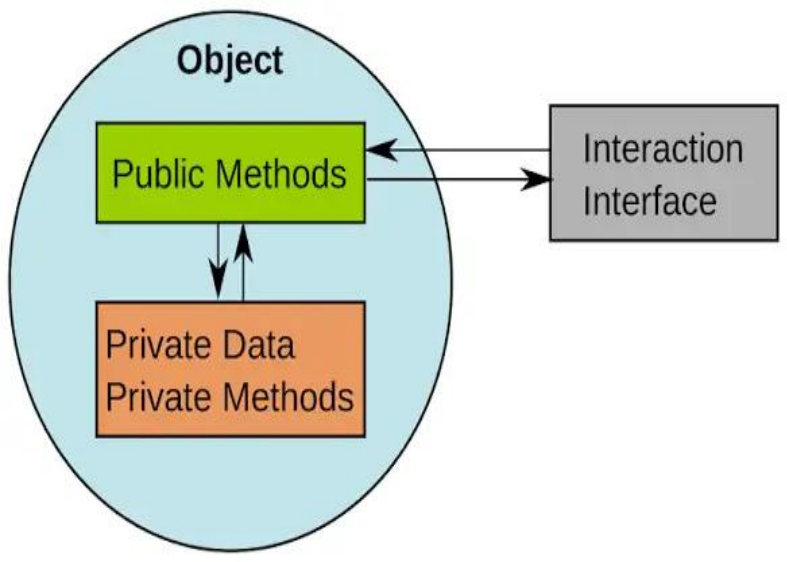




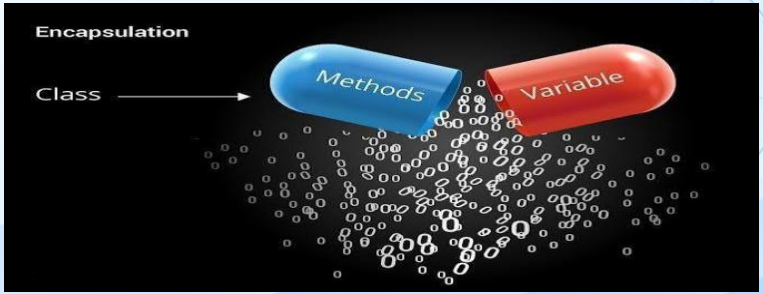


- **Transparency**
  Hide the dis.: access, location, concurrency, replication, failure
- **Consistency**
  Keep replicated data correct and up to date
- **Fault tolerance**
  Keep working despite node or network failures
- **Coordination**
  Order operations across independent nodes correctly

The Object Concept



- An object bundles data and the operations that act on it (encapsulation)
- Objects communicate by invoking each other's methods
- Foundation of object-oriented distributed systems (e.g. CORBA, RMI)

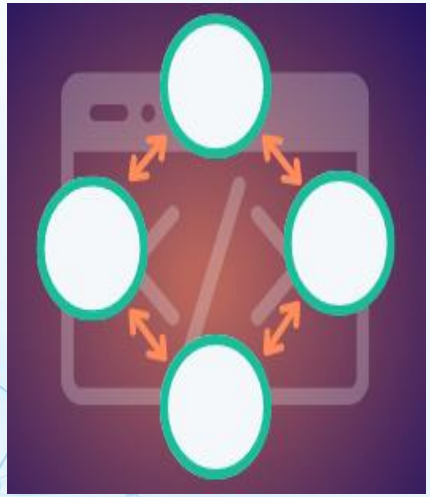


Object, Component and Service: three levels of granularity

Concept	Granularity	Exposes	Typical technology
Object	Fine-grained	Methods	Java objects, C++ classes
Component	Medium-grained	A reusable module with defined interfaces	EJB, .NET components
Service	Coarse-grained	A business function over a network	Web services, REST APIs

The cloud is built primarily on services: coarse-grained, network-accessible functions

Service-Oriented Architecture (SOA)



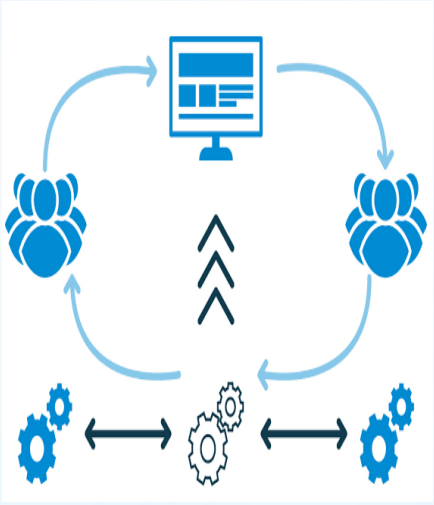
- **Loose coupling**

Services interact with minimal dependency on each other



- **Reusability**

A service can be used by many different applications



- **Interoperability**

Services communicate via standard protocols

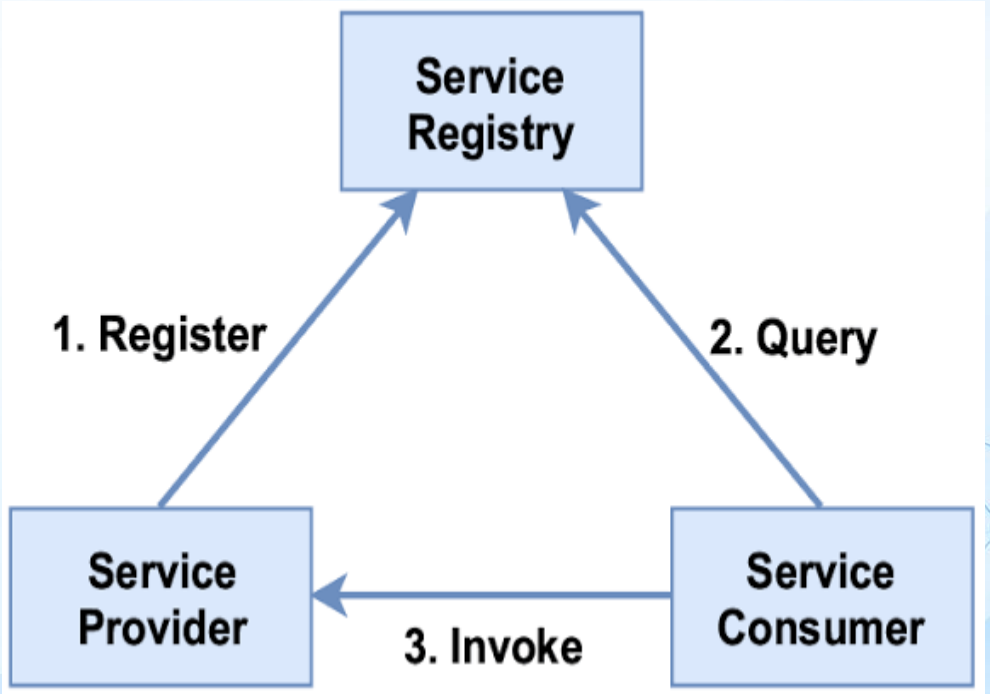


- **Discoverability**

Services can be found and invoked dynamically

SOA: the Three Roles

- **Provider** publishes a service description to the **registry**
- **Consumer** searches the registry, then binds directly to the **provider**
- *This dynamic discovery is a precursor of how cloud APIs and marketplaces work*

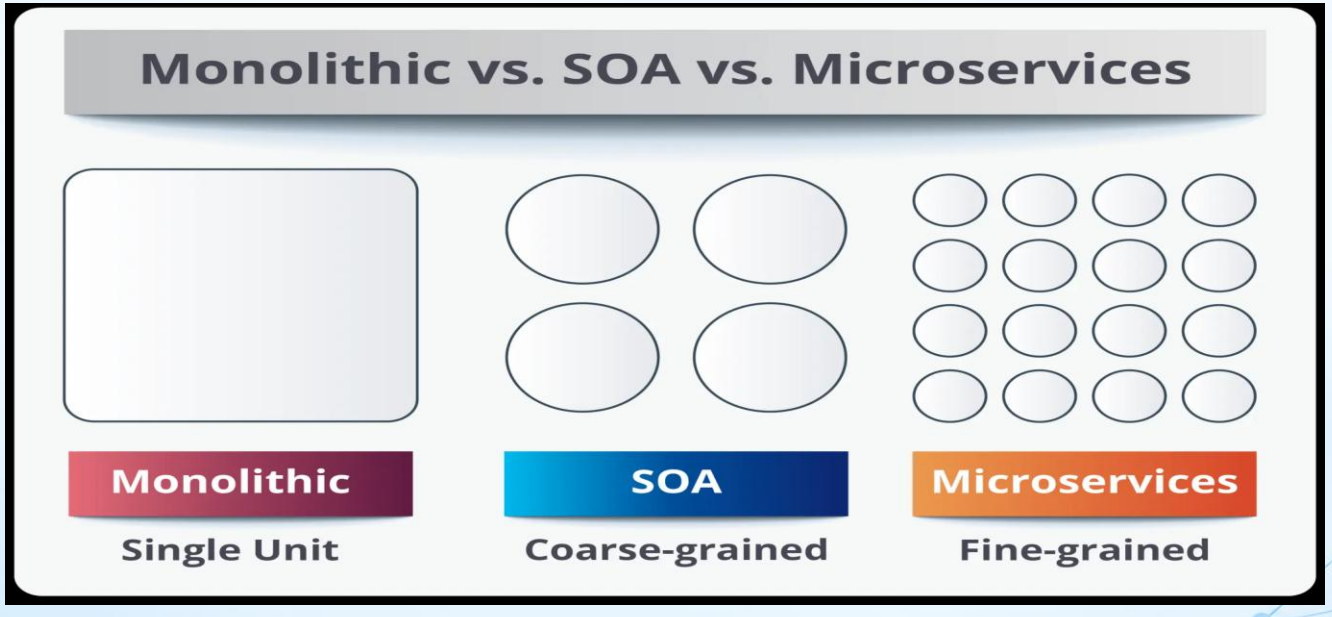


SOA vs Monolithic Architecture

Criterion	Monolithic	SOA
Structure	One large, tightly coupled application	Independent, loosely coupled services
Change impact	A change can affect the whole system	Services can change independently
Reuse	Difficult	High: services are shared across apps
Cloud relevance	Harder to scale piece by piece	Natural fit for scalable cloud services

SOA's ideas (loose coupling, services, APIs) directly shaped cloud and later microservices architectures

SOA vs Monolithic Architecture



SOA's ideas (loose coupling, services, APIs) directly shaped cloud and later microservices architectures

2. Cloud Computing: Definitions



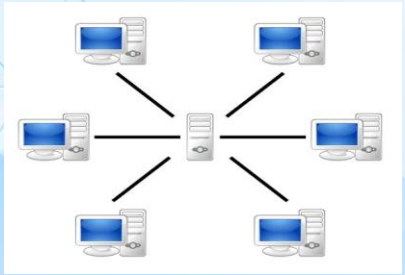
From distributed computing to a global utility

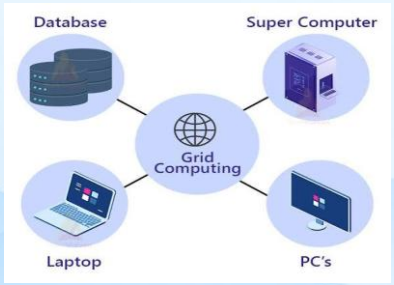
From Distributed Systems to the Cloud

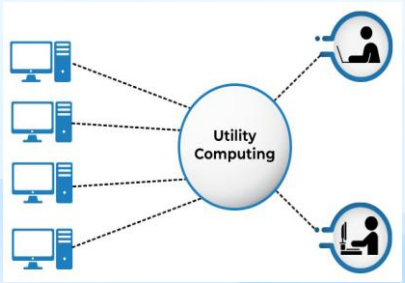
Canvas

Distributed Systems 1970s-1990s → **Grid Computing** shared computing power → **Utility Computing** pay-per-use resources → **Cloud Computing** today

- Each step added abstraction, self-service and elasticity
- The cloud packages decades of distributed-systems research into an accessible service





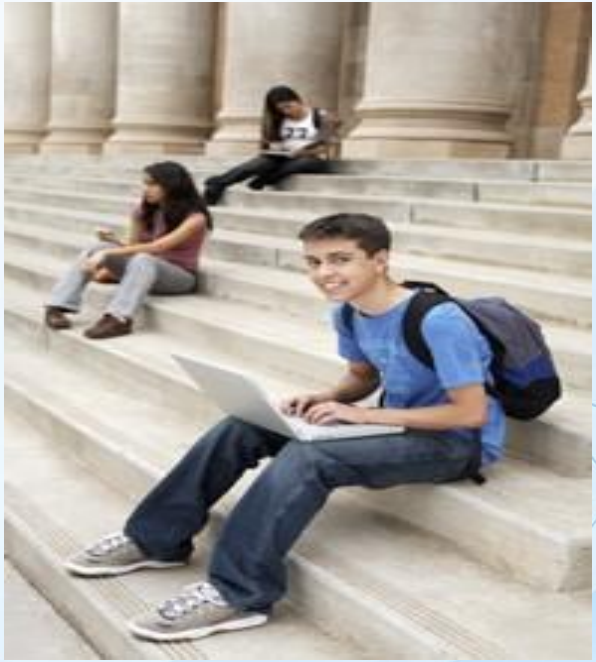




Context

- **Scenario: Development of an Online E-Commerce Website**

A computer science student, passionate about web development, has decided to create an online e-commerce website to sell products. The goal is to offer an intuitive, fast, and scalable platform capable of attracting customers.

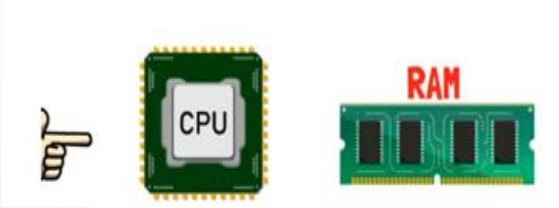


Context

Selected Technologies:

- **Backend**: PHP with the Laravel framework
- **Database**: PostgreSQL with PostGIS for managing geospatial data
- **Frontend**: JavaScript with Angular 4

Context







Offline site web

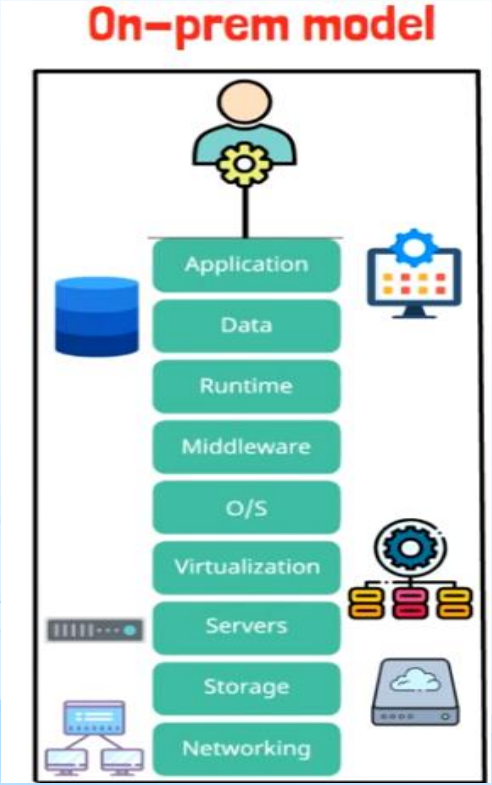


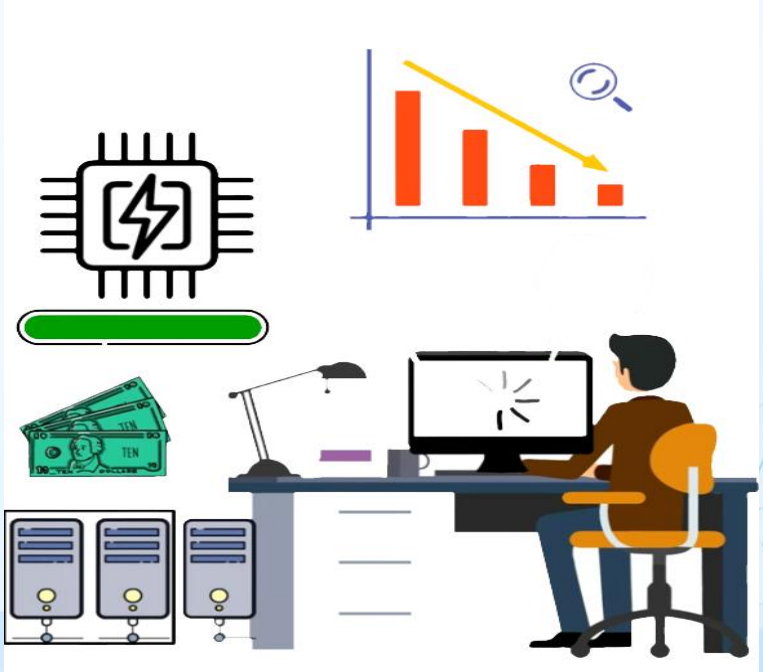
Ali

Black Friday



Context


On–prem model
Application
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking



Introduction

On –premise computing
On-prem computing

- Information technology is now essential, not only for individuals but also for businesses.
- All companies have many computers and several servers available to their employees. Computing tools and devices are becoming increasingly powerful and efficient.



Introduction

Corporate IT has become a tool for improving a company's *profitability* and *productivity*.

However, a single IT failure in your company can put you at risk of losing all your software data, or even a simple bug in your management software can bring your entire business to a standstill.



How can an organization’s IT department be managed?

Solution:



Cloud computing can provide protection against data loss in the event of an IT failure.



Cloud Computing (CC)

What are your initial thoughts when hearing “Cloud Computing”?

1- On-Demand Access

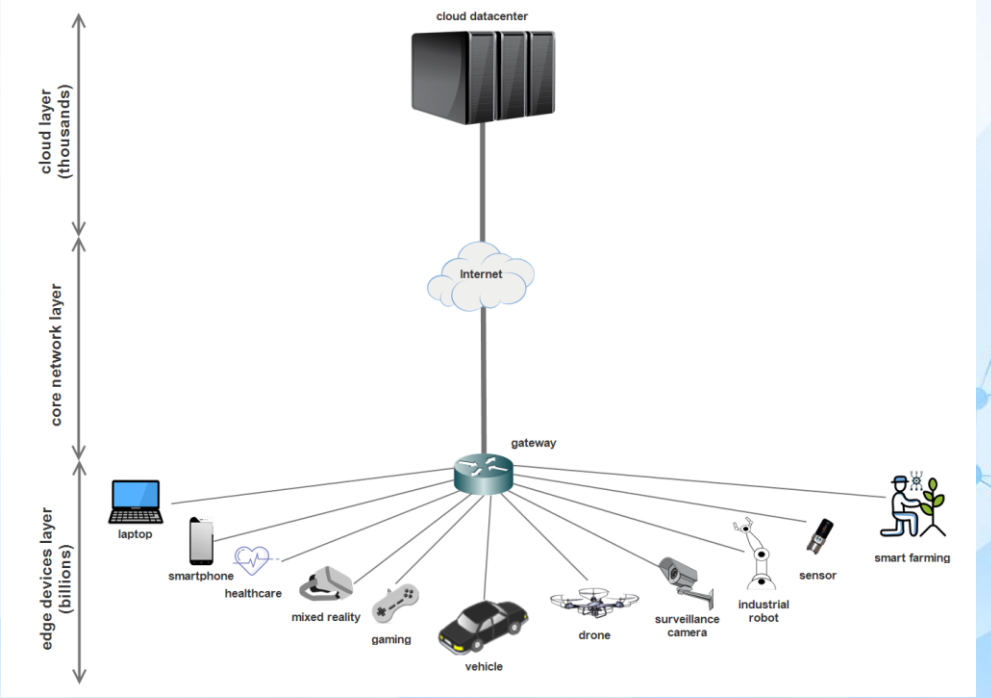


2- Scalability & Flexibility



3- Pay for What You Use





Defining Cloud Computing (CC)

What is cloud computing?



- Cloud computing is the **provision of IT services** (such as software, databases, servers, and networks) over the Internet.

- This means that end users can access software and applications regardless of where they are located.

Cloud Computing (CC)

- By outsourcing these resources, companies can access the IT resources they need so what they need, and when they need, without having to purchase or maintain physical on-site IT infrastructure.
- This solution ensures flexible resources, faster innovation, and economies of scale.



Defining Cloud Computing (CC)

Cloud Computing (CC) is the **on-demand delivery** of **computing resources / services (storage, processing, applications)** over the internet, allowing users to **access**, **scale**, and **pay** only for what they use without owning physical infrastructure.

- **Resources**: servers, storage, networks, applications, services
- « **Shared pool** »: many customers draw from the same physical infrastructure
- « **Rapidly provisioned** »: minutes, not weeks

NIST definition (paraphrased): a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort.

Virtual Spaces in Cloud Computing context

1- CC creates a **virtual space** where computing resources—like **servers**, **storage**, and **applications** exist **online** rather than on a **physical device**.

2- Everything happens **remotely** in **data centers / distant servers**.

Example:
When you save a file in **Google Drive**, it is not stored on **your computer** but in a **virtual space** that you can access **from anywhere**.

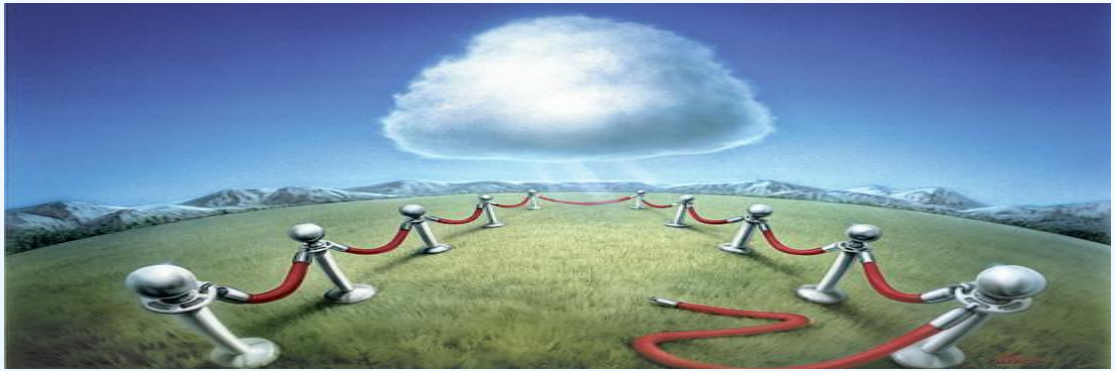
The Utility Computing Analogy

Electricity grid	Cloud computing
You don't own a power plant	You don't own data centers
You plug in and consume power	You call an API and consume compute
You pay per kWh used	You pay per hour / per request used
The provider manages generation	The provider manages infrastructure

Traditional IT vs Cloud IT

Criterion	Traditional IT	Cloud IT
Upfront cost	High (buy hardware)	Low (pay-as-you-go)
Provisioning time	Weeks	Minutes
Scaling	Manual, slow	Automatic, fast (elasticity)
Maintenance	Your team	Largely the provider

Historical Background of CC



1) 1950s: Mainframes & Time-Sharing

2) 1970s-90s: Virtual Machines & Hypervisors

3) 2000s: Cloud Computing Emerges

Historical Background of CC

1) 1950s – Large-Scale Mainframes & Time-Sharing

Computing was expensive, so multiple users shared a single machine.

a) ***Mainframes***: Large, powerful computers used by many users.

b) ***Time-sharing***: Users accessed the same CPU & storage in turns (efficient but limited).

c) ***Example***: IBM mainframes in businesses & universities.

Historical Background of CC

2) 1970s-90s – The Evolution of Virtual Machines (VMs)

🚩 **Problem**: Each mainframe could only run one OS at a time.

💯 **Solution**: Virtualization allowed multiple VMs on a single machine.

a) *VM Concept*: Software-based computers that run independently on shared hardware.

b) *Benefits*: Improved resource utilization, isolation, and flexibility.

c) *Example*: IBM's VM/370 OS (1972).

Historical Background of CC

The Emergence of Hypervisors

⚙️ **Definition**: A Hypervisor is a software that manages multiple VMs on one physical server.

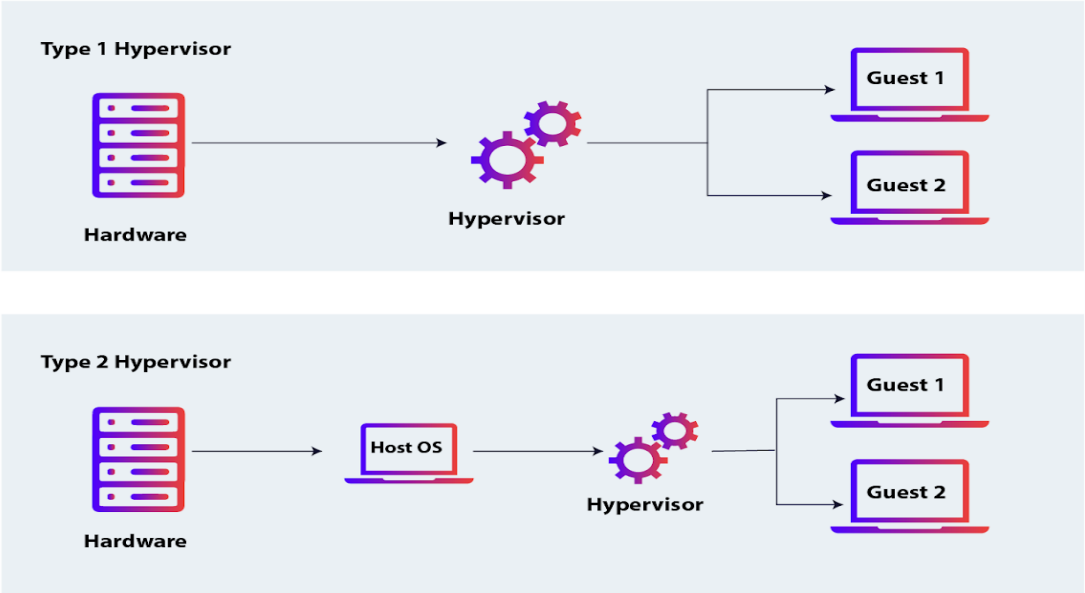
Types: Bare-metal (Type 1) vs. Hosted (Type 2).

- 1970s: IBM develops first hypervisors.
- 1990s-2000s: VMware popularizes virtualization in enterprise computing.

💡 **Impact**: Hypervisors paved the way for cloud computing by enabling efficient resource sharing.

Historical Background of CC

Hypervisor types



Historical Background of CC

When Did Clouds First Emerge?

- 1990s: Telecommunication companies started offering virtualized computing resources.
- 1999: Salesforce.com launched Software as a Service (SaaS) (first cloud-like service).
- 2006: Amazon Web Services (AWS) introduced Elastic Compute Cloud (EC2) → true cloud era begins.

💡 **Takeaway**: Cloud computing became possible due to VMs & hypervisors, but it truly emerged in the early 2000s.

Historical Background of CC

3) Early 2000s – The Cloud Computing Era

🚀 **The Shift to Cloud Services:**

- 2006: AWS launches EC2, offering scalable computing power online.
- 2008: Google & Microsoft enter cloud computing with Google App_Engine & Azure.
- 2009: Cloud computing gains popularity with public, private, and hybrid models.

💡 **Key Impact**: Businesses moved from buying servers to renting scalable resources.

3. Characteristics of Cloud Computing



The 5 essential traits

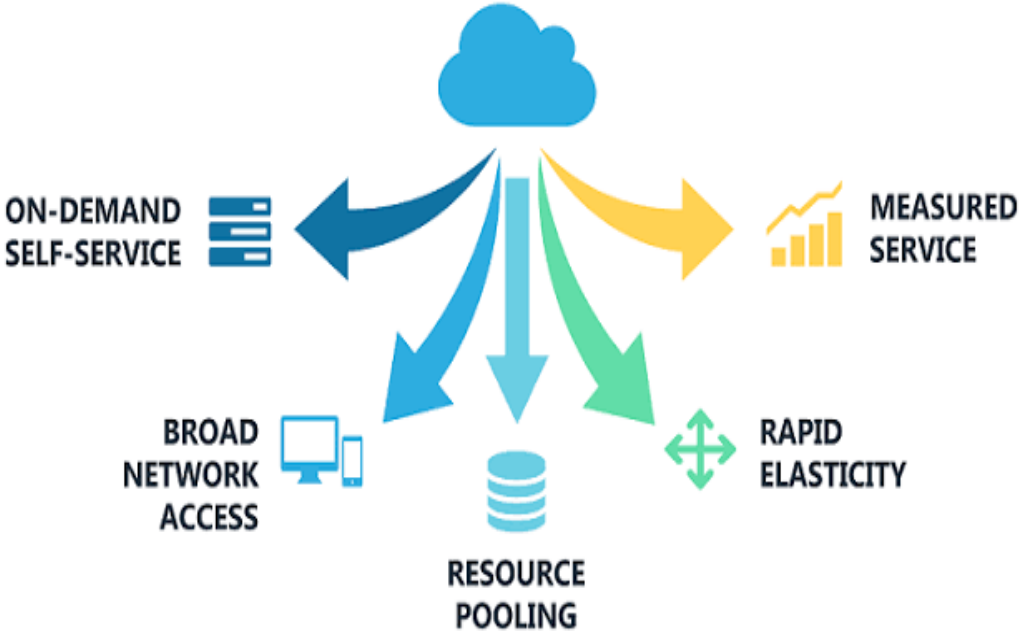
The 5 Essential Characteristics (NIST)

- **On-demand self-service**

Provision resources without human interaction with the provider

- **Broad network access**

Available over the network, from various devices



- **Rapid Measured service**

Usage is monitored, controlled and billed

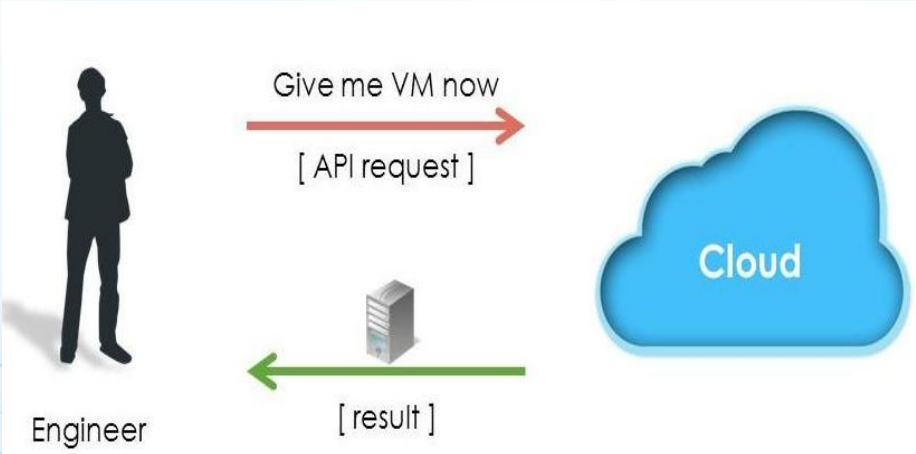
- **Rapid elasticity**

Resources scale up or down automatically

- **Resource pooling**

Multi-tenant model shares physical resources

On-demand self-service



- No phone call, no ticket, no waiting for an administrator
- This is what makes cloud fundamentally faster than traditional IT procurement

User
clicks / calls API

→

Cloud console or API
provisions resource

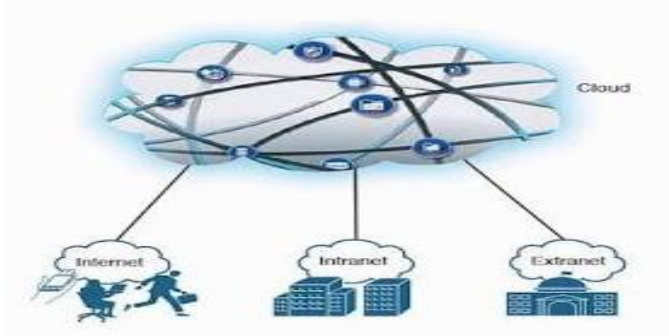
→

Resource ready
in minutes, no human involved

Broad Network Access and Resource Pooling

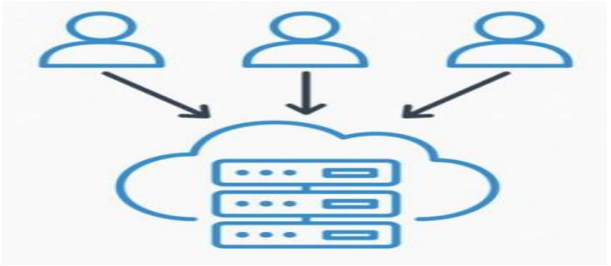
Broad network access

- Services are reachable over standard networks (the internet)
- Accessible from laptops, phones, tablets, servers
- Standard protocols: HTTPS, APIs



Resource pooling

- Physical resources serve multiple customers (multi-tenancy)
- Customers usually don't know the exact physical location
- Enables high utilization and lower cost per customer



Rapid Elasticity



- Elasticity works both ways: scale out and scale back in
- To the customer, capacity often appears unlimited

Low demand few resources active → **Traffic spike** auto-scaling triggers → **High demand** more resources added automatically → **Demand drops** resources released

Measured Service

Resource	Common billing unit
Compute (VM)	Per second / per hour of usage
Storage	Per GB stored per month
Data transfer	Per GB transferred out
API calls	Per thousand / million requests

Metering enables transparent billing and lets customers optimize cost

4. Deployment Models



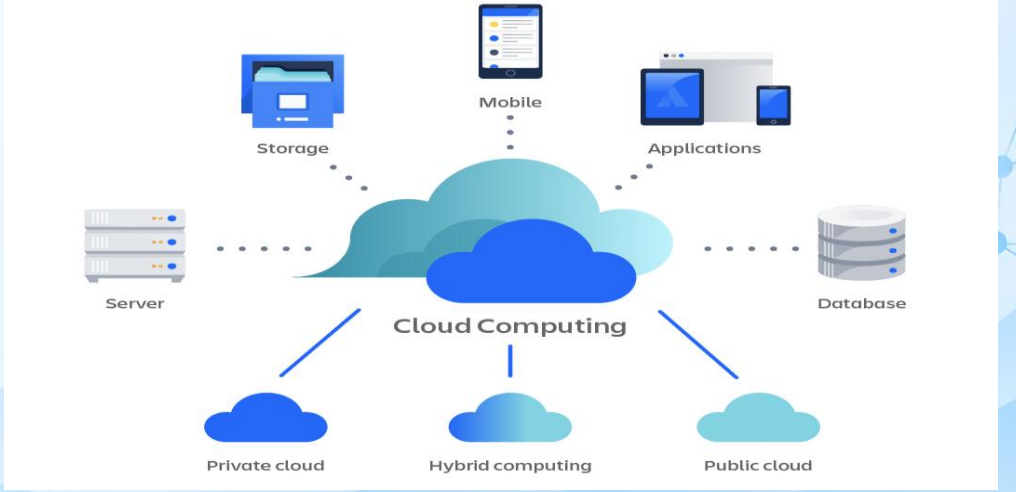
Public, private, hybrid, community

Cloud Models

Cloud models (or **deployment models)** define where **cloud resources are hosted** and **who manages them**.

Four main models:

- **Public Cloud**
- **Private Cloud**
- **Hybrid Cloud**



Cloud Models



Examples

e.g., AWS, Google Cloud, Microsoft Azure

1) Public Cloud = Public Bus 🚍

Principle: A **public bus** is like a **public cloud**—it's shared by multiple users, operated by a third party, and available on demand.

✅ **Pros**:

- Low cost (pay per ride).
- No maintenance required (managed by the transport provider).
- Easily scalable (more buses/routes can be added).

❌ **Cons**:

- Less control over schedule and route
- Security risks (shared space, less privacy)
- Performance varies depending on passenger load

Cloud Models



2) Private Cloud = Personal Car 🚙

e.g., On-premise VMware Private Cloud, OpenStack, Azure Stack

Principle: **Owning a private car** is like having a **private cloud**—dedicated resources, full control, but higher costs.

✅ **Pros**:

- Full control over routes, timing, and security
- Better privacy (no sharing with strangers)
- Customizable (can modify features as needed)

❌ **Cons**:

- High upfront costs (buying and maintaining the car)
- Requires management (fuel, repairs, parking)
- Scaling is expensive (buying more cars)

Cloud Models



e.g., AWS Outposts, Google Anthos, Microsoft Azure Hybrid

3) Hybrid Cloud = Taxi 🚕

Principle: A **taxi or ride-sharing service** is like a **hybrid cloud**—you combine ownership (private car) with on-demand access (public transport).

✅ **Pros**:

- Call a taxi when needed.
- Cost-effective: Pay only when additional transport is required.
- Balanced security: Keep sensitive travel plans private.

❌ **Cons**:

- More complex to manage (deciding when to use private vs. public transport)
- Can get costly (frequent taxi use adds up)
- Coordination needed (ensuring smooth transitions between private & public options)

Cloud Models



❖ Examples (cont.)

4) Community Cloud = Carpooling 🚋

e.g., Government & Healthcare Clouds

Principle: **Carpooling** with colleagues or friends is like a **community cloud**—a shared system among users with common needs.

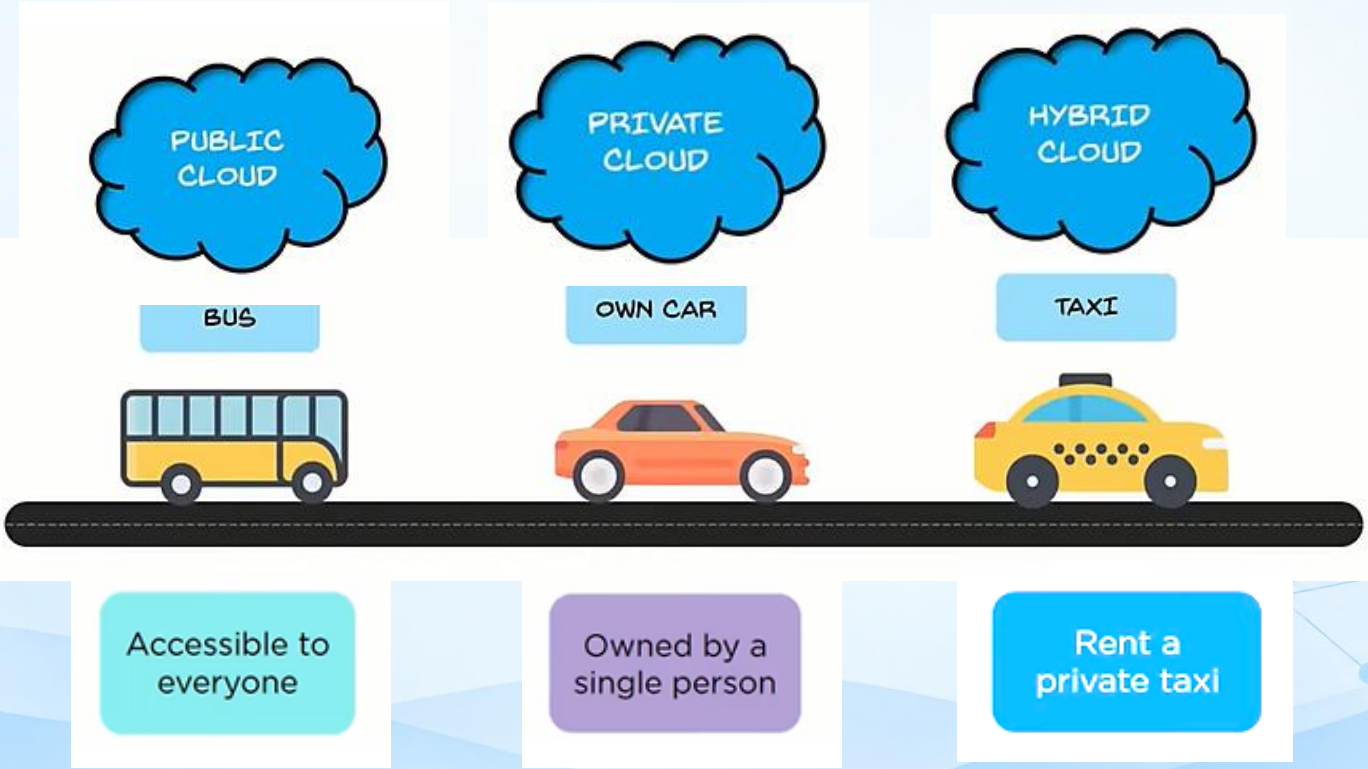
✅ Pros:

- Cost-sharing among members.
- More control than public transport (can choose routes, timings).
- Meets common needs (same workplace, same schedule).

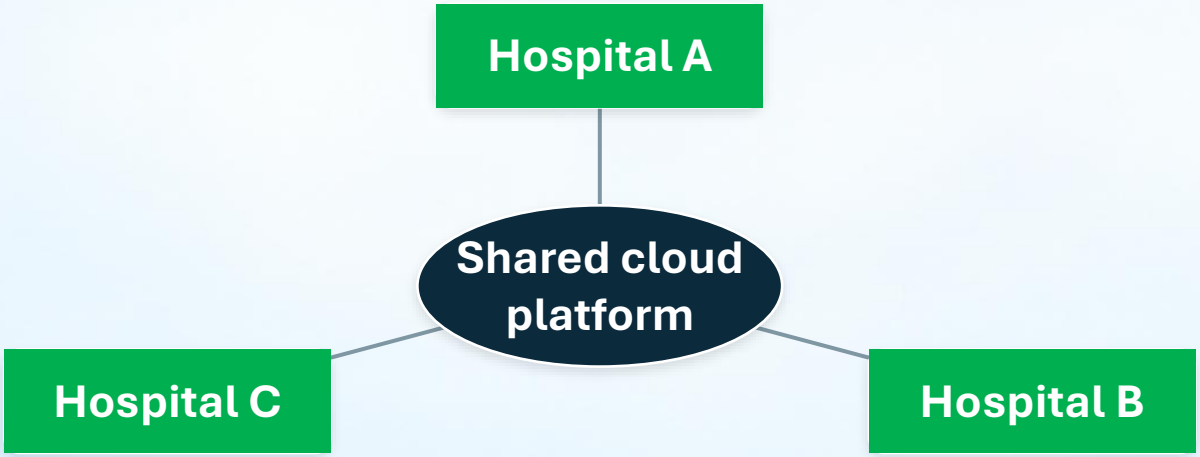
❌ Cons:

- Limited scalability (fixed number of seats).
- Governance challenges (who manages the carpool, what if someone is late?).
- Not as widely available (specific to a group).

Cloud Models



Community Cloud



Several organizations with similar requirements (e.g. compliance, sector) share the same infrastructure and costs

Cloud Models

Definitions

Public Cloud → Services are *provided over the internet* and *shared by multiple users*. **Low cost**, **high scalability**, but **less control**.



Private Cloud → Dedicated to a *single organization*, offering **more security** and **control** but at a **higher cost**.



Hybrid Cloud → Mix of *public and private clouds*. It brings **flexibility** and **cost-efficient** but introduces **complex management**.



Community Cloud → Shared by organizations with similar needs (e.g., government, healthcare), providing **shared costs** and **security compliance** but with **limited scalability**.



Cloud Models

Summary

Cloud Model	Transport Example	Ownership	Cost	Flexibility	Scalability
Public	Public Bus	Shared	Low	Low	High
Private	Personal Car	Single User	High	High	Low
Hybrid	Taxi/Uber + Own Car	Mixed	Medium	High	High
Community	Carpooling	Group-Owned	Medium	Medium	Low

💡 **Conclusion**: The exact cloud model is chosen based on the scenario, use case and specific needs.

5. Cloud Services



IaaS, PaaS, SaaS and beyond

Cloud Services

Cloud services are **computing resources** delivered over the internet, encompassing **IaaS** (infrastructure), **PaaS** (platforms), and **SaaS** (software), while **on-premise** services are managed within an organization's infrastructure.

Four main types of services:

1) On-Premises

2) IaaS (Infrastructure as a Service)

3) PaaS (Platform as a Service)

4) SaaS (Software as a Service)



Affected components:

Applications	Middleware	Servers
Data	OS	Storage
Runtime	Virtualization	Networking

Cloud Services





1) On-Premises

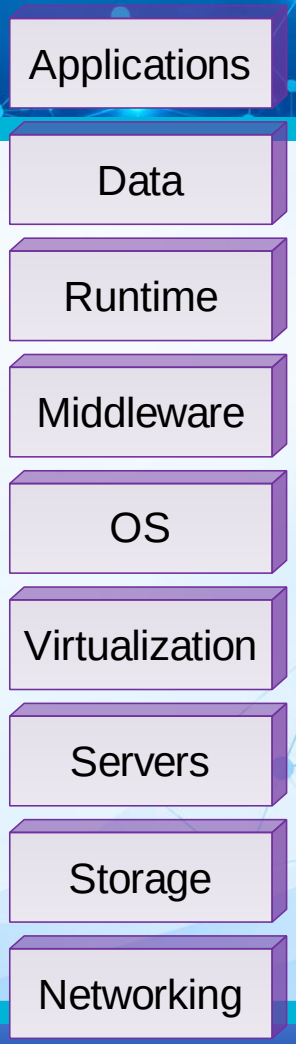
On-premises refers to **IT infrastructure**, **software**, and **data** that are *hosted* and *managed* within an organization's own **physical facilities**, rather than relying on third-party cloud providers.

Who manages it?

YOU

Others

P.S. “On-Premises” is a “*general* service model” but not a “*cloud* service model”



Cloud Services

1) On-Premises

✅ **Pros**:

- **Full Control** → Organizations manage hardware, software, and security.
- **Better Security** → Sensitive data is not exposed to third-party providers.
- **No Dependency on Internet** → Systems work without cloud connectivity.

❌ **Cons**:

- **High Cost** → Requires large investments in hardware, maintenance, and IT staff.
- **Limited Scalability** → Expanding requires purchasing new hardware.
- **Complex Management** → IT teams must handle everything from security to updates.

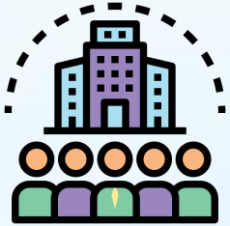
Cloud Services

1) On-Premises Examples

- Banks & Financial Institutions



- Government Data Centers



- Hospitals & Healthcare Systems



Cloud Services

2) IaaS (Infrastructure as a Service)

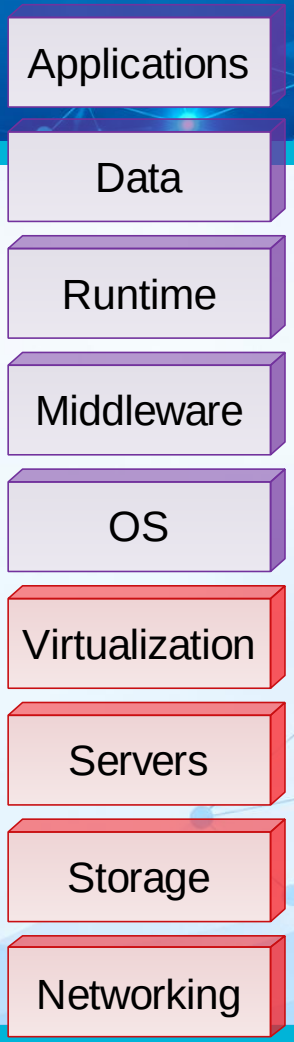
IaaS provides **virtualized computing resources** *over the internet*, including **servers**, **storage**, **networking**, and **virtualization**. Organizations can scale their IT infrastructure without owning or managing physical hardware.

Who manages it?

YOU

Others





Cloud Services

2) IaaS (Infrastructure as a Service)

✅ **Pros**:

- **Scalability** → Quickly scale resources up or down as needed.
- **Cost Savings** → Pay only for what you use, no need to buy hardware.
- **Disaster Recovery** → Cloud providers offer high availability and backups.

❌ **Cons**:

- **Security Concerns** → Data is stored in third-party data centers.
- **Internet Dependency** → Requires a stable internet connection to access.
- **Management Complexity** → Users still need to configure and maintain servers.

Cloud Services

2) IaaS Examples

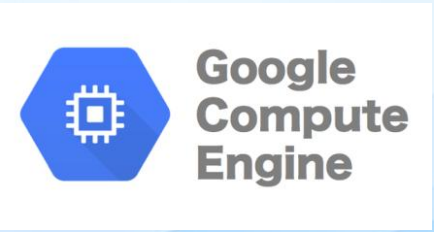
- Amazon Web Services (AWS EC2, S3)



- Microsoft Azure VMs



- Google Cloud Compute Engine



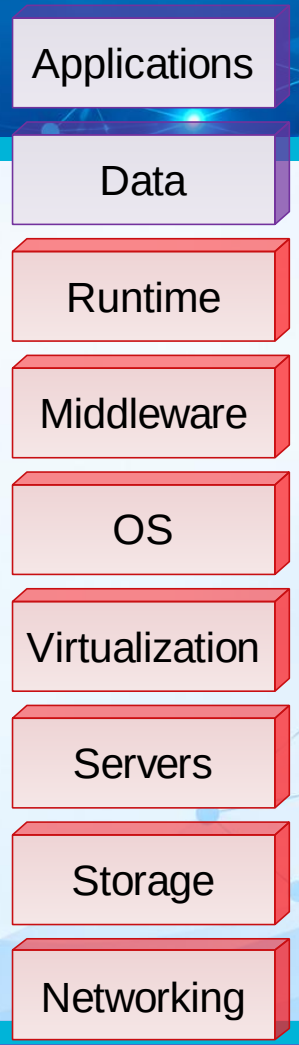
Cloud Services

3) PaaS (Platform as a Service)

PaaS provides a **cloud-based platform** for *developing*, *testing*, and *deploying applications*. It includes **tools**, **databases**, **middleware**, and **development frameworks**, allowing developers to focus on *coding* rather than *infrastructure management*.

Who manages it?

- YOU
- Others



Cloud Services

3) PaaS (Platform as a Service)

✅ **Pros**:

- **Faster Development** → Developers can build apps without managing infrastructure.
- **Cost-Effective** → No need to invest in hardware or OS maintenance.
- **Automatic Scaling** → The platform adjusts resources based on demand.

❌ **Cons**:

- **Limited Customization** → Users depend on platform restrictions and configurations.
- **Vendor Lock-In** → Switching providers can be difficult.
- **Security Risks** → Data security depends on the provider's policies.

Cloud Services

3) PaaS Examples

- Google App Engine



- Microsoft Azure App Services



- Heroku



Cloud Services

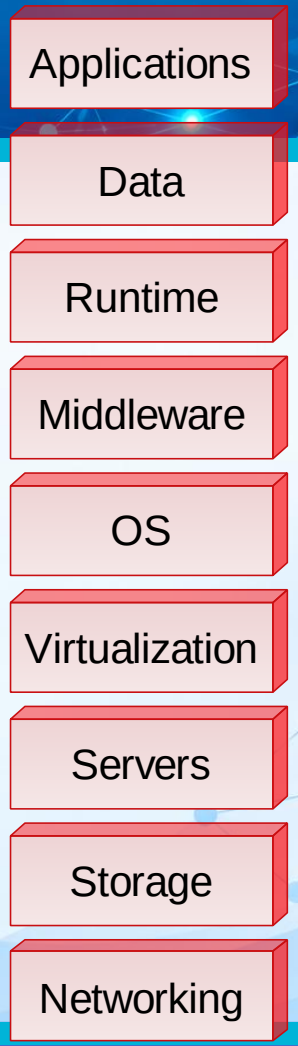
4) SaaS (Software as a Service)

SaaS delivers **fully functional software applications** *over the internet*, removing the need for *installation*, *maintenance*, or *infrastructure management*. Users simply access the service via **a web browser** or **app**.

Who manages it?

- YOU
- Others





Cloud Services

4) SaaS (Software as a Service)

✅ **Pros**:

- **Easy Access** → Use from anywhere with an internet connection.
- **Automatic Updates** → No need to install or update software manually.
- **Lower Upfront Costs** → Pay a subscription instead of buying software.

❌ **Cons**:

- **Limited Control** → Users depend on provider updates and features.
- **Internet Dependency** → No access if there's no connection.
- **Security & Privacy Concerns** → Data is stored and managed by a third party.

Cloud Services

4) SaaS Examples

- Google Workspace
(Docs, Gmail, Drive)



- Microsoft 365
(Word, Excel, Outlook)

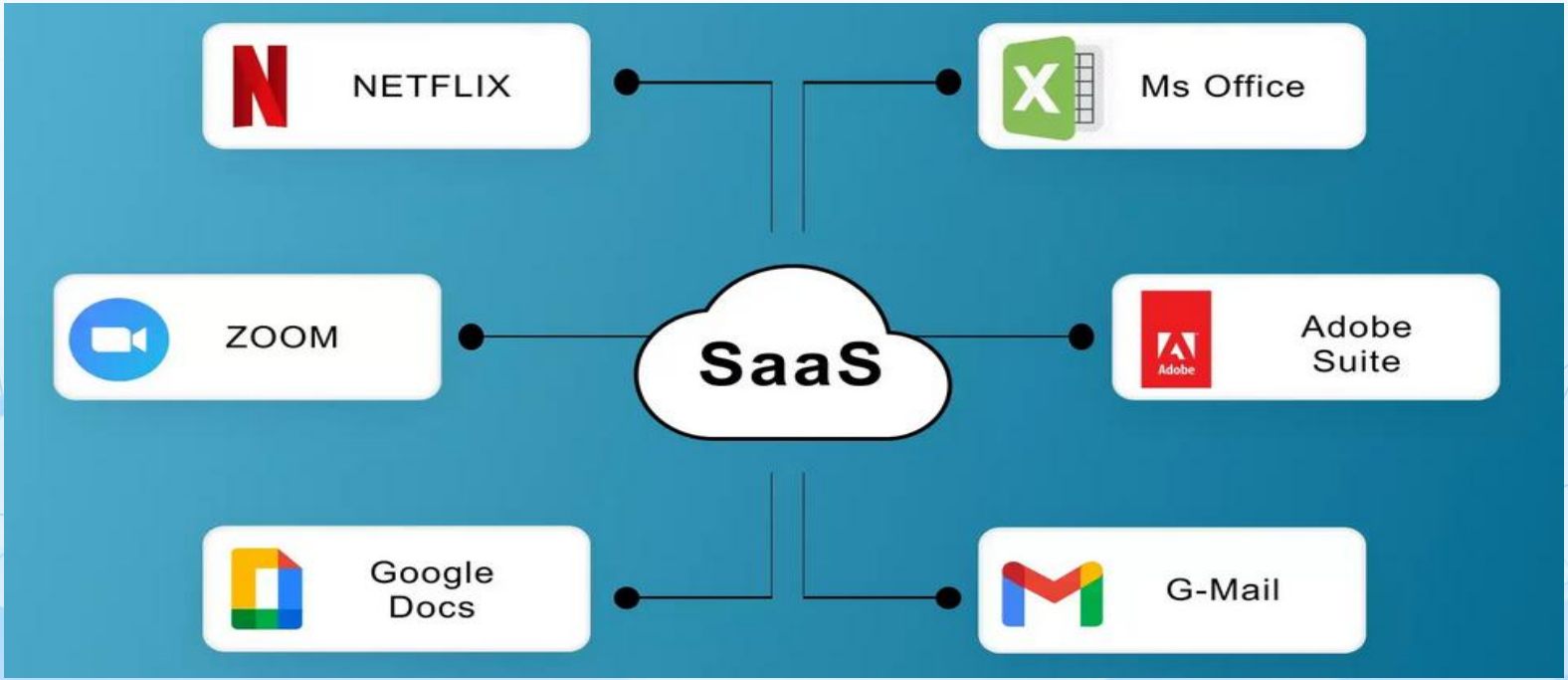


- Salesforce



Cloud Services

4) SaaS Examples



Top 25 Software As A Service Examples


Momo Board



Salesforce


Dropbox


Microsoft


Shopify


Slack


FreshBooks


Toggl Track


Paychex


Google Workspace


Intercom



Xero


Evernote


Zendesk


Figma



RingCentral


Sketch


GoToConnect



WooCommerce


HelloSign


Loom

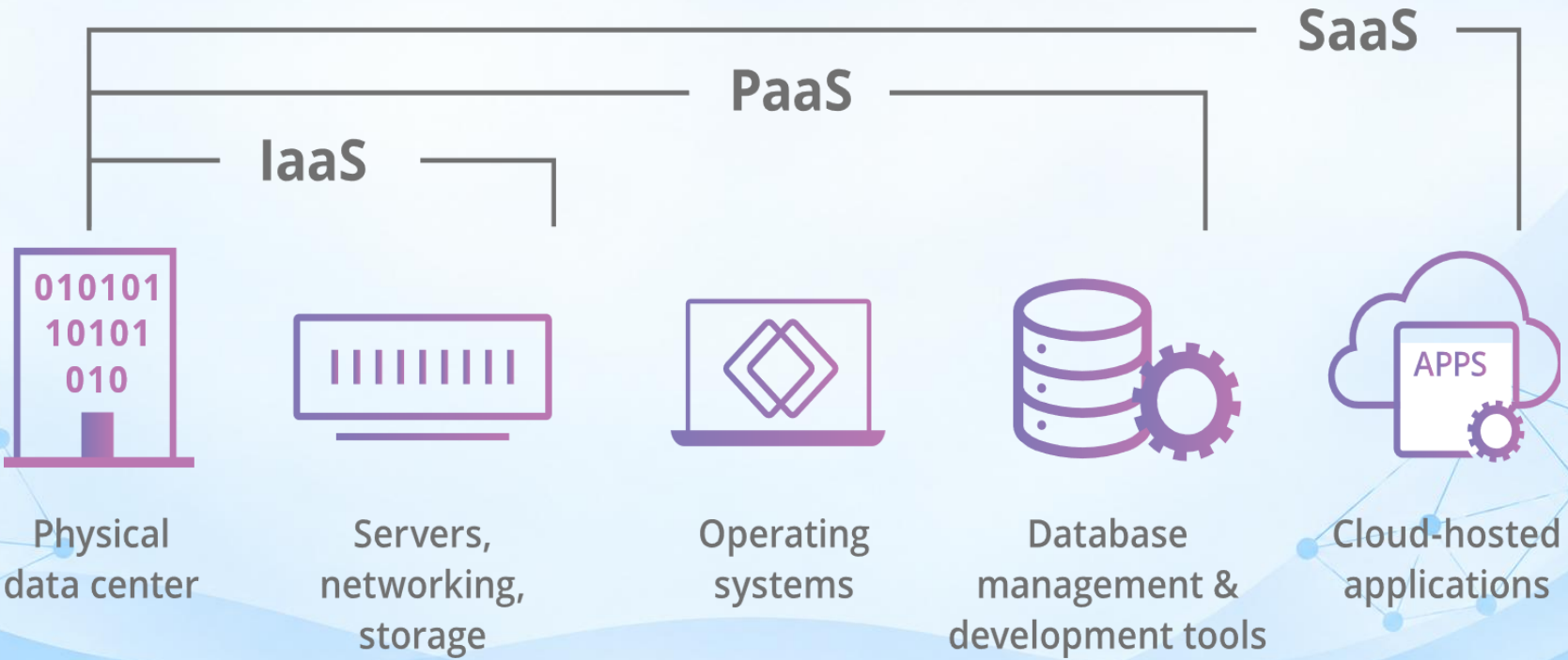

Whimsical


Mailchimp

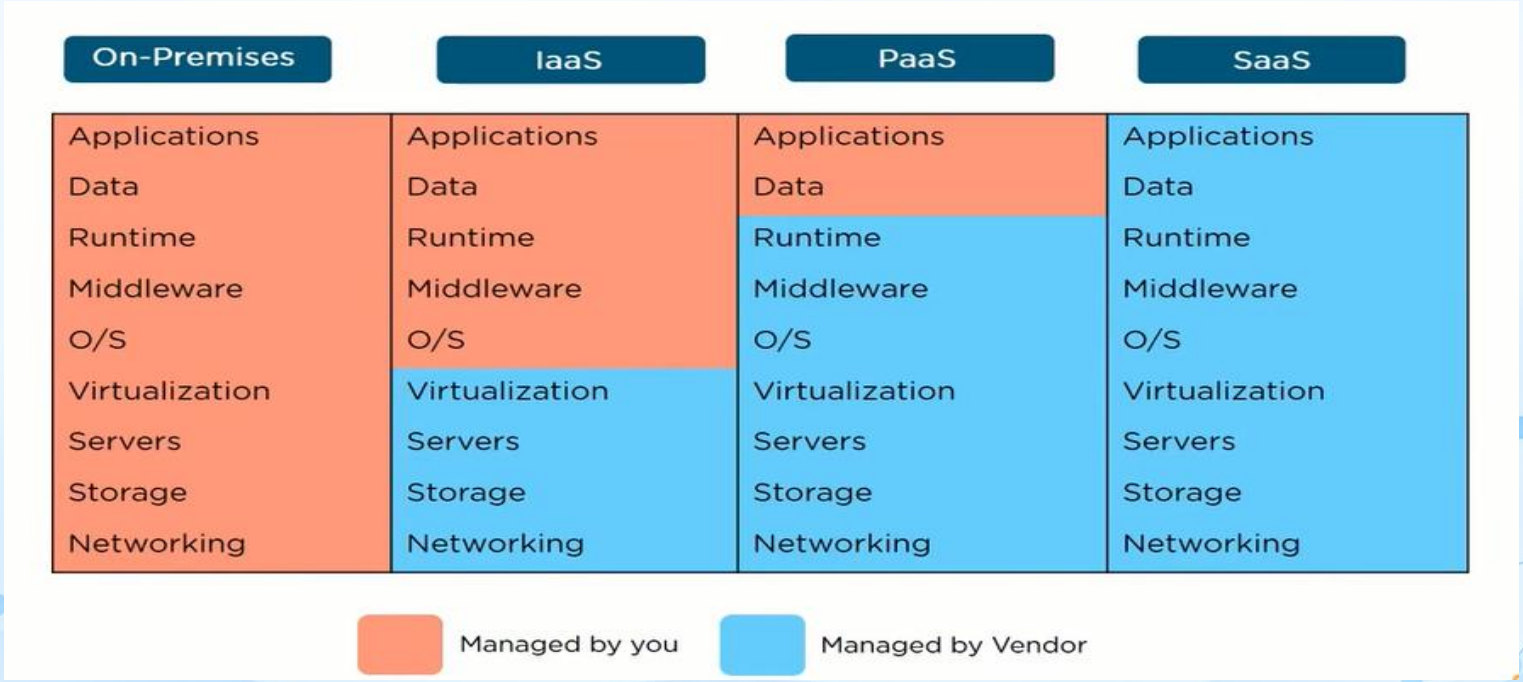

Kissmetrics


Kajabi

Types of cloud services (SaaS, PaaS, IaaS)

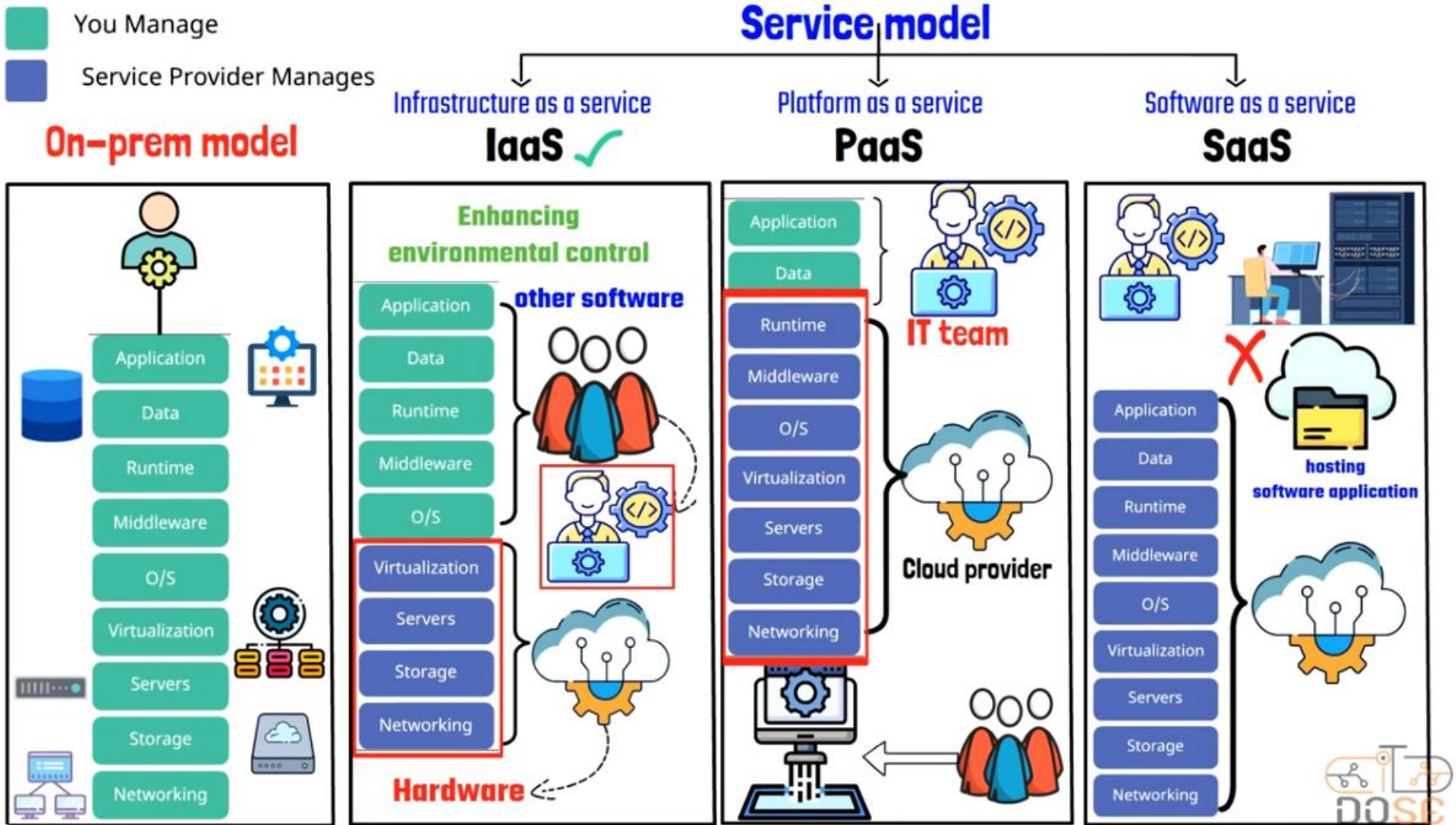


Types of cloud services (SaaS, PaaS, IaaS)



Middleware : Middleware software that enables applications to communicate and work together.

Runtime : The execution environment (for example, Java Virtual Machine, .NET) required to run the application's code.


You Manage
Service Provider Manages
Service model
Infrastructure as a service
IaaS
Platform as a service
PaaS
Software as a service
SaaS
On-prem model
Application
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking
Enhancing environmental control
other software
Hardware
IT team
Cloud provider
hosting software application

Cloud Services

Which cloud service is suitable for you?



IAAS

If your business needs a virtual machine, opt for Infrastructure as a Service



PAAS

If your company requires a platform for building software products, pick Platform as a Service



SAAS

If your business doesn't want to maintain any IT equipment, then choose Software as a Service



6. Edge Computing & Fog Computing

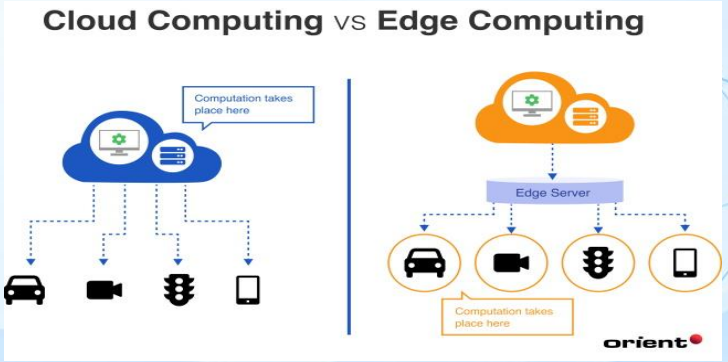


Definitions and Differences

Edge Computing

Edge computing is an architecture in which data is processed and stored as close as possible to its source. This can be done on smart devices, sensors, or edge servers.

- Edge computing is often used for applications that require **ultra-low latency** or **real-time decision-making**, such as **augmented reality**, **virtual reality**, **and autonomous vehicles**.



Edge Computing

With the rise of **IoT (Internet of Things)**, connected devices are becoming more and more numerous.

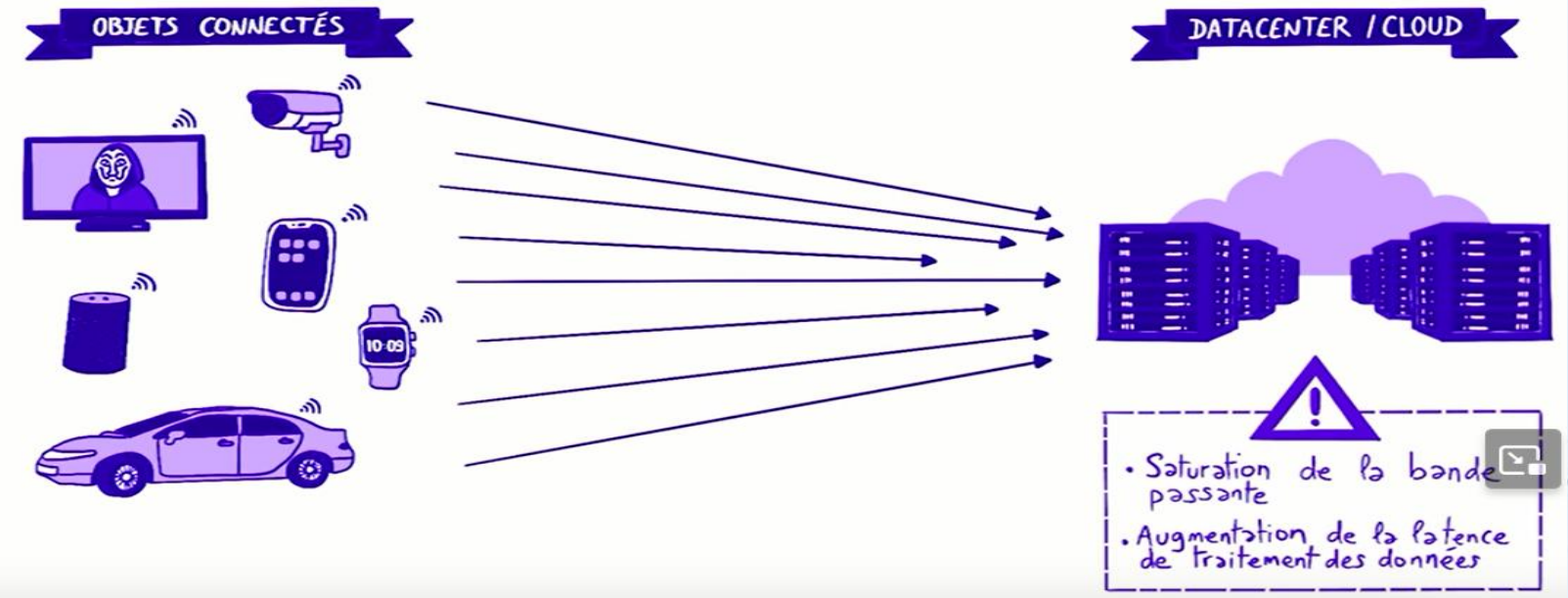
For example:

Smartwatches, smart cars, smart TVs, connected surveillance systems, personal assistants, smartphones...

All of this means that the **volume of data generated is rapidly increasing**. This **massive data volume** risks **saturating the bandwidth** between these connected devices and the **data centers** that process the data.

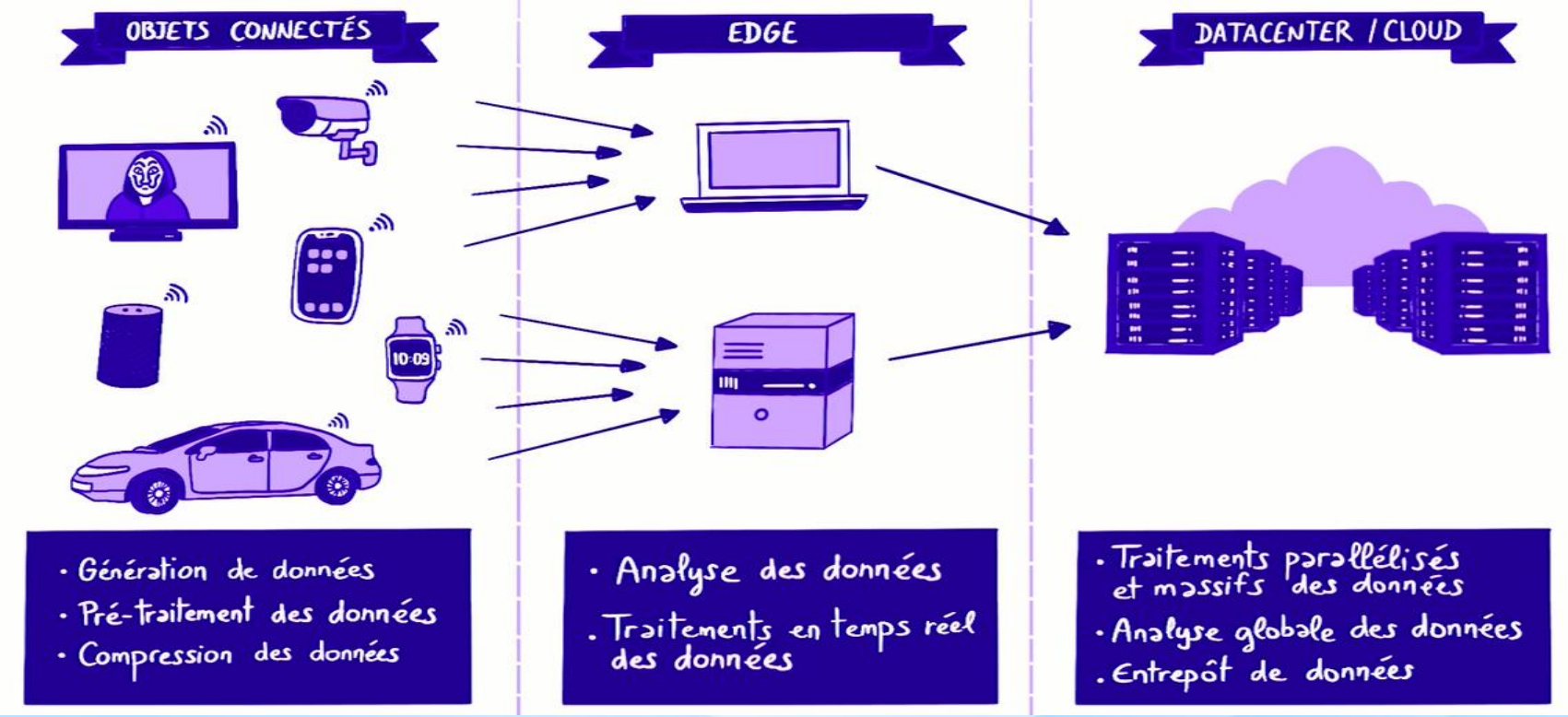


Edge Computing



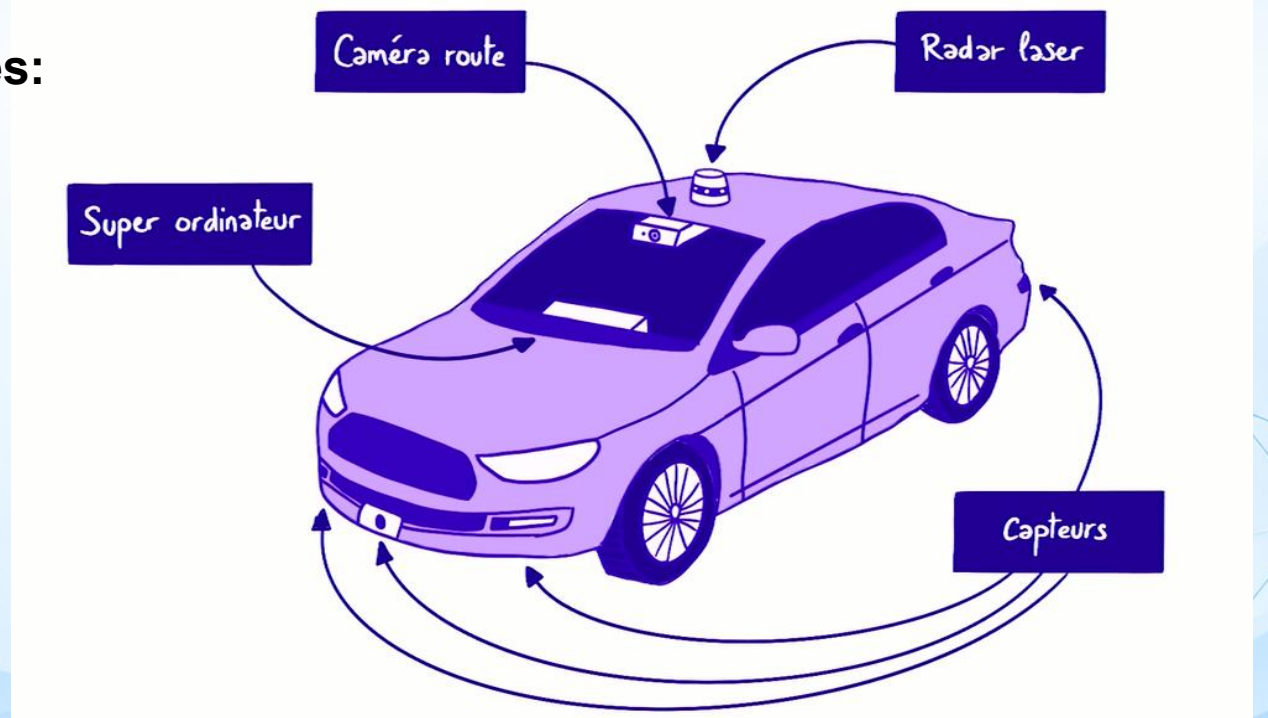
The solution offered by **Edge Computing** is simply to **process part of these large volumes of data in a decentralized way at the local level**.

Edge Computing



Edge Computing

Autonomous vehicles:

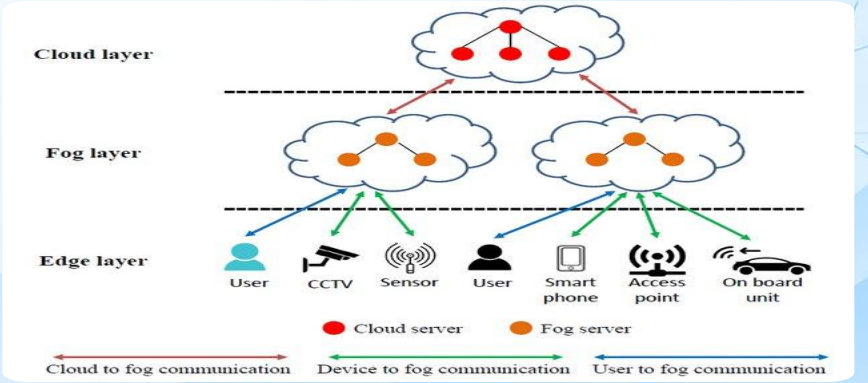


Fog Computing

Overview

Fog computing is an architecture in which data is **processed and stored in a node located between the cloud and edge devices**. This node can be a **router** or a **switch**.

- Fog computing is often used for applications that require **lower latency than the cloud**, but **not as low as edge computing** such as **video surveillance and online gaming**.



Difference between Edge&Fog Computing

Difference

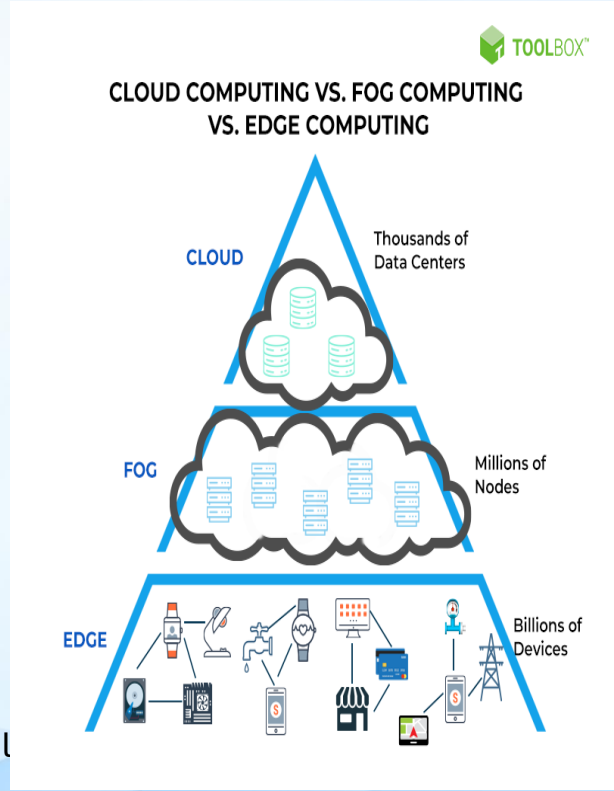
Edge computing focuses on **local data processing close to edge devices**, while **Fog computing** **extends** this idea by creating an **intermediate computing layer** closer to edge devices, but still at some distance from the central cloud.

Both approaches aim to **improve efficiency**, **responsiveness**, **and** **data management** in distributed environments.

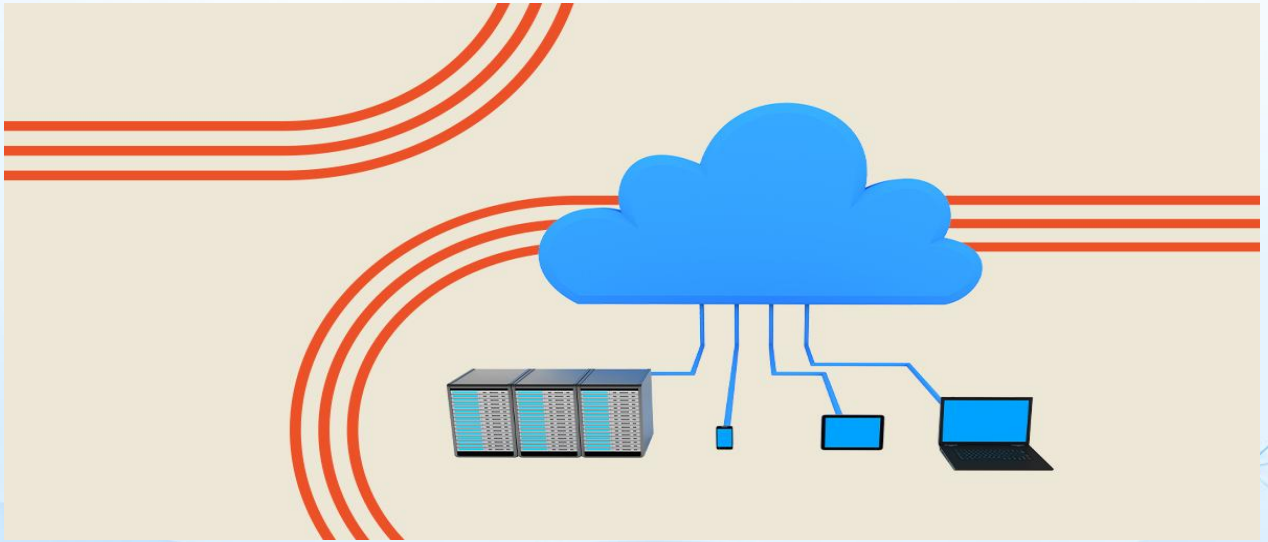
Difference between Edge&Fog Computing

Difference

Feature	Edge Computing	Fog Computing
Processing location	On the device itself or very close to it	On intermediate nodes between device and cloud (e.g., routers, switches)
Latency	Ultra-low (fastest response)	Low, but higher than edge
Data handling	Handles data from individual devices	Aggregates and processes data from multiple devices
Use cases	Autonomous vehicles, AR/VR, smart cameras	Video surveillance, online gaming, industrial IoT



7. Cloud Computing Infrastructure



Definitions, Iot, Difference

Cloud Computing IT Infrastructure

Some Related Concepts

- PC vs. Workstation vs. Server?





Q- Can they be mobile?



PCs are built for affordability and general use.

Workstations are "power-user PCs" for demanding tasks.

Servers are built for multi-user environments and for reliability, uptime and high networking.

Cloud Computing Infrastructure

Feature	PC (Personal Computer)	Workstation	Server
Purpose	General use (web, office)	Heavy apps (design, 3D, data)	Runs services for other systems
Performance	Basic to moderate	High-end CPU, GPU, RAM	Optimized for uptime & load
Hardware	Consumer-grade	Professional-grade	Server-grade (ECC RAM, RAID)
Users	1 user	1 power user	Many simultaneous users
OS	Windows/macOS/Linux	Windows Pro/Linux	Server OS (e.g., Ubuntu Server)
Examples	Home laptop or desktop	AutoCAD, 3D rendering stations	Web, database, file servers

Cloud Computing IT Infrastructure

- **Server** (Physically)



Tower Server



Rack Server



Blade Server

Cloud Computing IT Infrastructure



Type	Format	Ideal for	Cost	Density
Rack	Horizontal (19")	Businesses / Data centers	Medium	High
Blade	Modules in chassis	Large data centers	Very high	Very high
Tower	Like a PC	Small and medium businesses	Low to medium	Low

Cloud Computing IT Infrastructure

❑ Server (Logically)

- Serving users with several different **types of services**, based on **the needs**.



Cloud Computing IT Infrastructure



- **Rack**

  - A metal **frame** or **cabinet** that holds **multiple servers** and components in a **vertical stack**.

  - Standard size:

    + 19 inches (≈ 0.483 meters) wide.

    + measured in U, 1U = 1.75 inches (≈ 0.044 meters) height.

  - Common rack sizes: **42U**, **48U**.

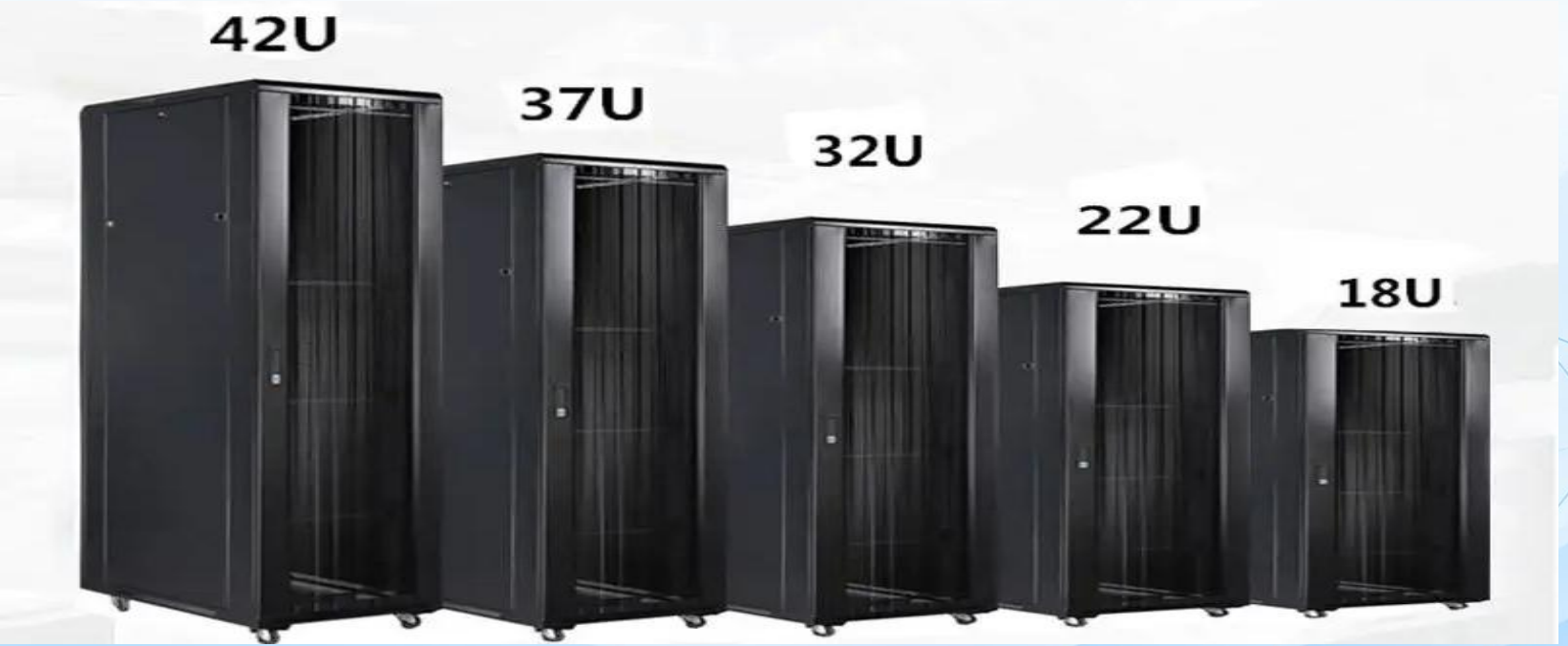
- **Rack Unit (U)**

  - A unit of measurement for **server height**.

  - 1U, 2U, etc. – defines how much **vertical space** a device occupies **in a rack**.

Cloud Computing IT Infrastructure

Rack Example:



Cloud Computing IT Infrastructure

- **Rack Server**

A **server** that fits into **a rack**
usually 1U to 4U.

Designed for **density**
and **easy maintenance**.

Example:



Cloud Computing IT Infrastructure



- **Chassis / Enclosure**

  The **housing unit** for multiple **blade servers** or other modular hardware.

  Provides shared resources like **power**, **cooling**, and **networking**.

- **Blade Server**

  A very thin **modular server** that slides into a **blade enclosure**.

  Exploits shared **power**, **cooling**, **networking**, making it **space** and **energy efficient**.

Cloud Computing IT Infrastructure

Blade Server & Chassis

Example:



Cloud Computing IT Infrastructure

- **Tower Server**

Are **standalone** servers, used in **small offices** or **environments with few servers.**

Example:

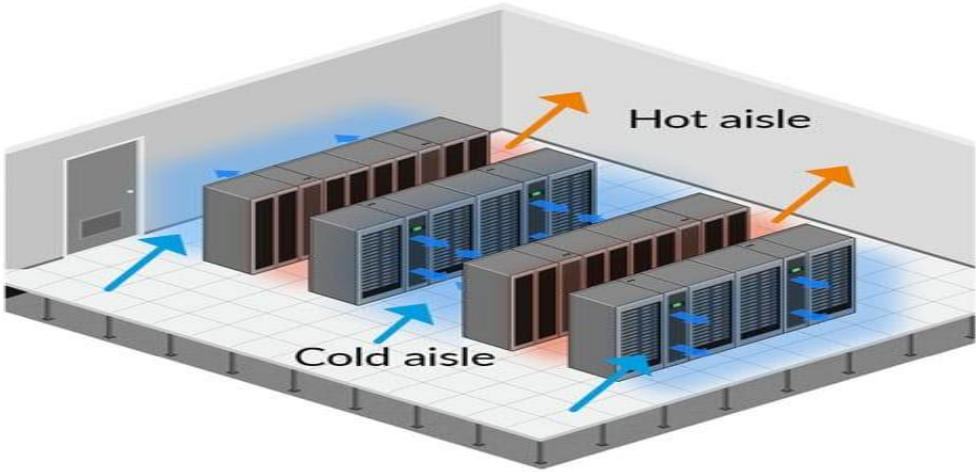


Cloud Computing IT Infrastructure

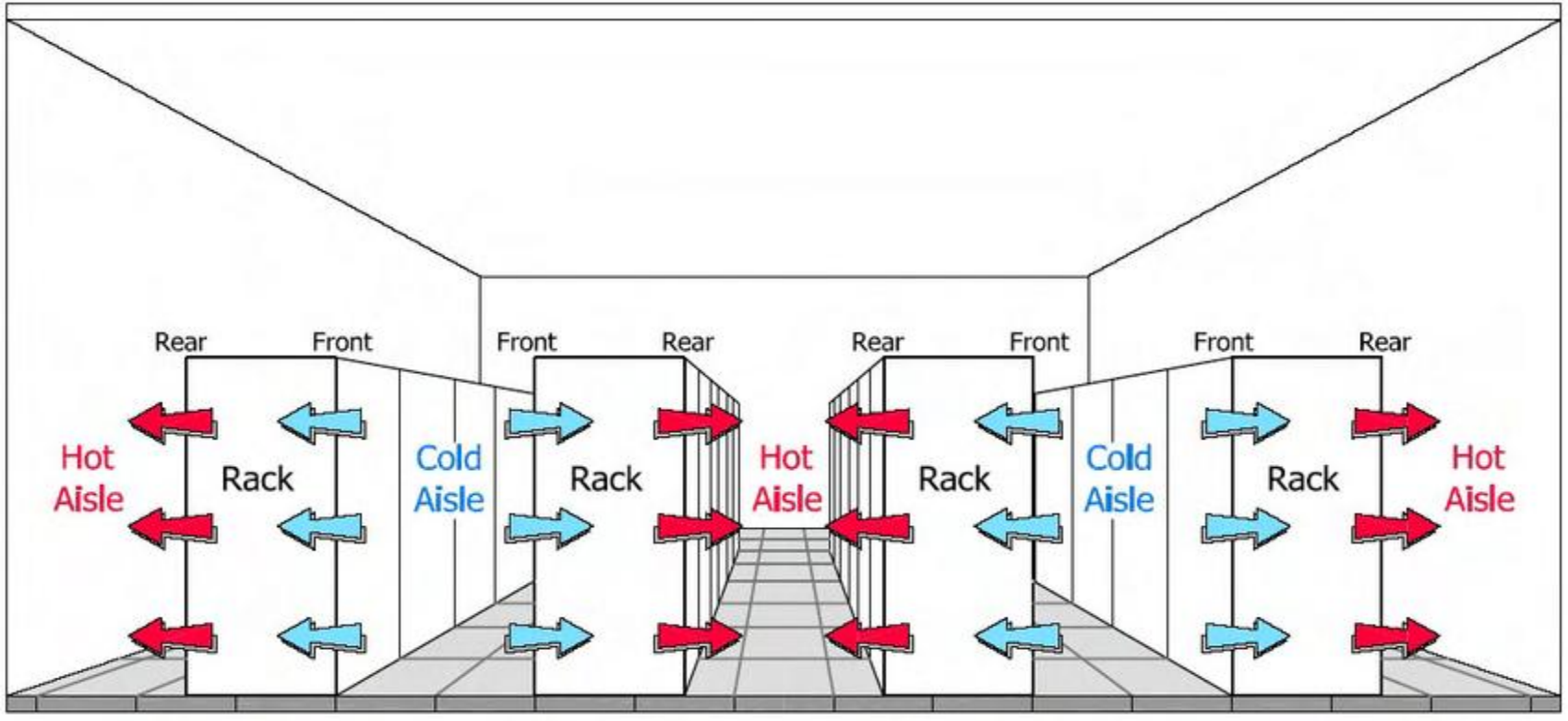
❑ Cooling

- Datacenters need serious **cooling** to prevent **overheating**.

- Includes **air cooling**, or **liquid cooling**, **cold aisle**/**hot aisle** configurations.



Cloud Computing IT Infrastructure



Cloud Computing IT Infrastructure

❑ Power Supply (UPS, PDU)

- **UPS** (Uninterruptible Power Supply): **Keeps systems running** during **power loss**.



- **PDU** (Power Distribution Unit): **Distributes electricity** to devices in a rack.



Cloud Computing IT Infrastructure

❑ Networking Equipment 🌐

Includes **switches**, **routers**, **firewalls**.

Handles data traffic *inside* the **datacenter** and *outside* to the **internet**.



Switch



Routers



Firewall

Cloud Computing IT Infrastructure

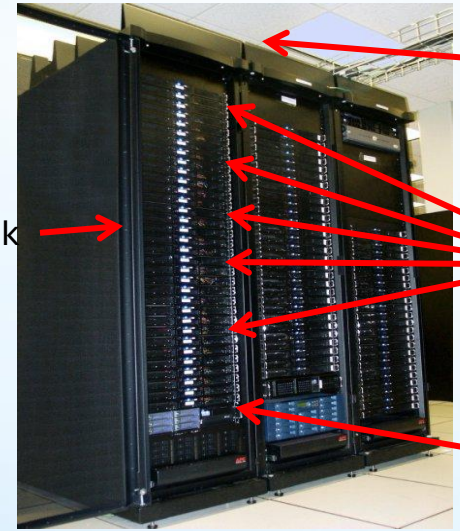
❑ Cable Management 🧰

- **Organizing** cables to avoid **clutter** and allow **airflow** and **maintenance**.

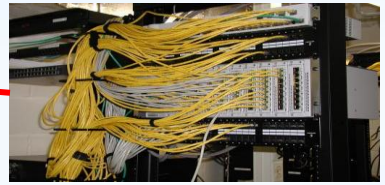




Cloud Computing IT Infrastructure



Rack



Network switch (connecte les nœuds entre eux et avec d'autres racks)



Plusieurs nœuds/blades (Souvent identiques)



Storage device(s)

Characteristics of a Cluster:

- Multiple similar machines, closely interconnected (same room?)
- Standardized hardware (racks)
- Generally **owned and used by a single organization**

Cloud Computing IT Infrastructure

Scaling up



PC



Server



Cluster



Data center

Solution: Data Center

Cloud Computing IT Infrastructure

Data Center: A site that hosts all the computer systems necessary for running IT applications.

A data center generally hosts a **large number of heterogeneous computer systems** interconnected through a network.





Cloud Computing IT Infrastructure

What does a data center look like?

Cooling plant

Data centers (size of a football field)





Google data center in The Dalles, Oregon

A computer the size of a **warehouse.**

A single data center can easily contain **10,000 racks** with **100 cores in each rack** (a total of **1,000,000 cores**).

Google's data center in Hamina, Finland, can hold **more than 12,000 racks**.

Cloud Computing Providers

Top Cloud Service Providers in 2026





Cloud Computing IT Infrastructure

Scaling up



PC



Server



Cluster



Data center



Network of data centers

And if a single data center is not enough?

Solution : Build additional data centers.

Questions to consider:

- **Where? (location)**
- **How many? (quantity)**

Cloud Computing IT Infrastructure

Data centers are generally distributed **all around the world**.





Example: The image above shows **Google data center locations**.

Why so many data centers?

Need to be **close to users** (physically!)

Cheaper resources

Protection against failures

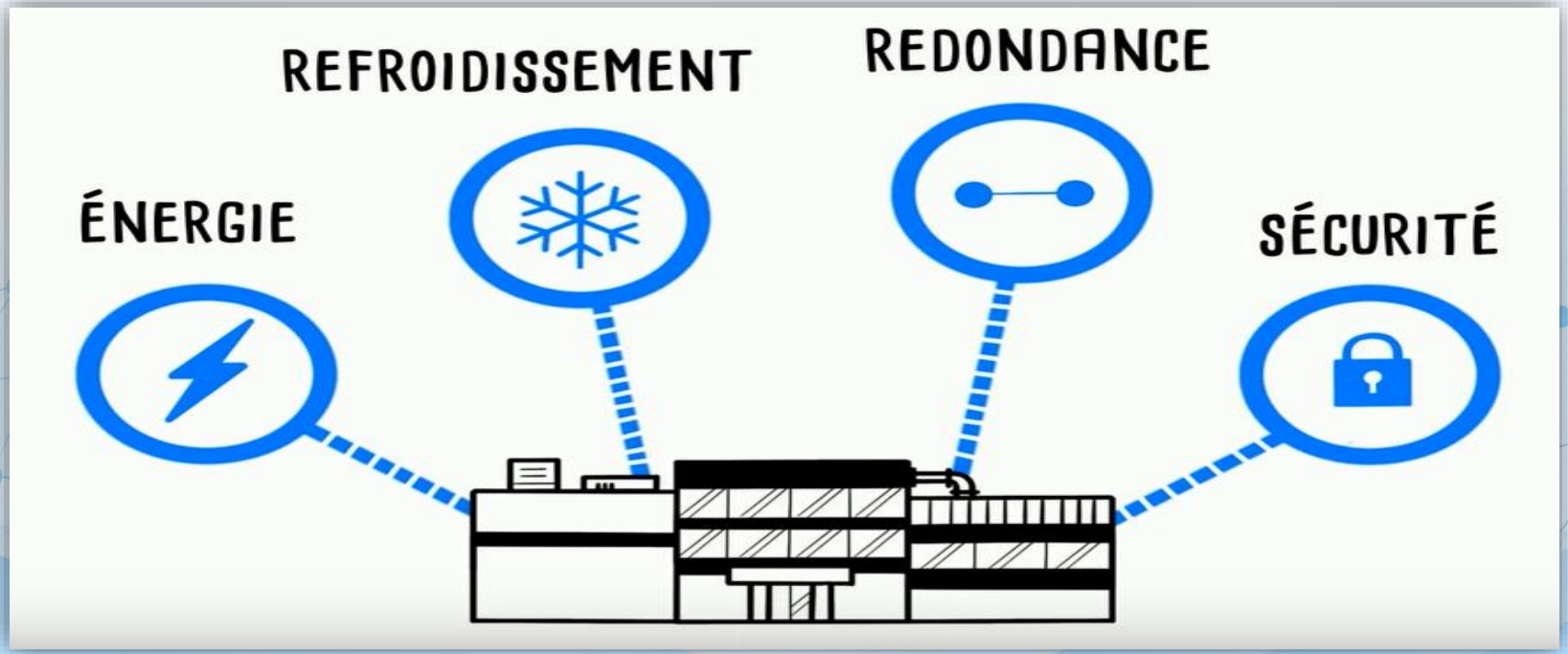
Cloud Computing IT Infrastructure

Amazon Web Services (AWS)



Cloud Computing IT Infrastructure

Datacenter Prerequisite

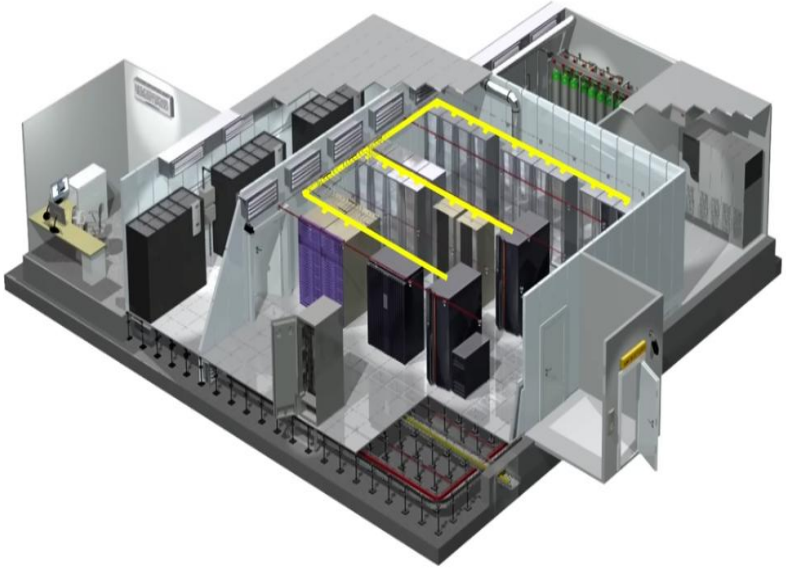


Cloud Computing IT Infrastructure

Elements of a Data Center

Non-IT Infrastructure:

Electrical transformers, power supplies, generato
Cooling cabinets (temperature, humidity control,
Fire protection, etc.



IT Infrastructure:

Servers, cabling, racks, network equipment
Storage, applications
System tools, etc.

Administration:

Operations personnel who **manage, maintain, and repair** both IT and non-IT systems.

Cloud Computing IT Infrastructure

Top three Largest Data Centers in the World

1. China Telecom-Inner Mongolia (Hohhot, China)

Power Capacity: 150 MW
Area: 994,062 square meters (10.6 million square feet)
Cost: $3 billion



2. The Citadel – Switch (Tahoe Reno, Nevada, USA)

Power Capacity: 650 MW
Area: 668,901 square meters (7.2 million square feet)
Cost: $1.5 billion



3. Yotta NM1 (Panvel, India)

Power Capacity: 50 MW
Area: 76,180 square meters (820,000 square feet)
Cost: over $1.1 billion



Cloud Computing IT Infrastructure

Examples



Silicon Valley Data Center Market



CoreSite VA3, Reston Virginia.



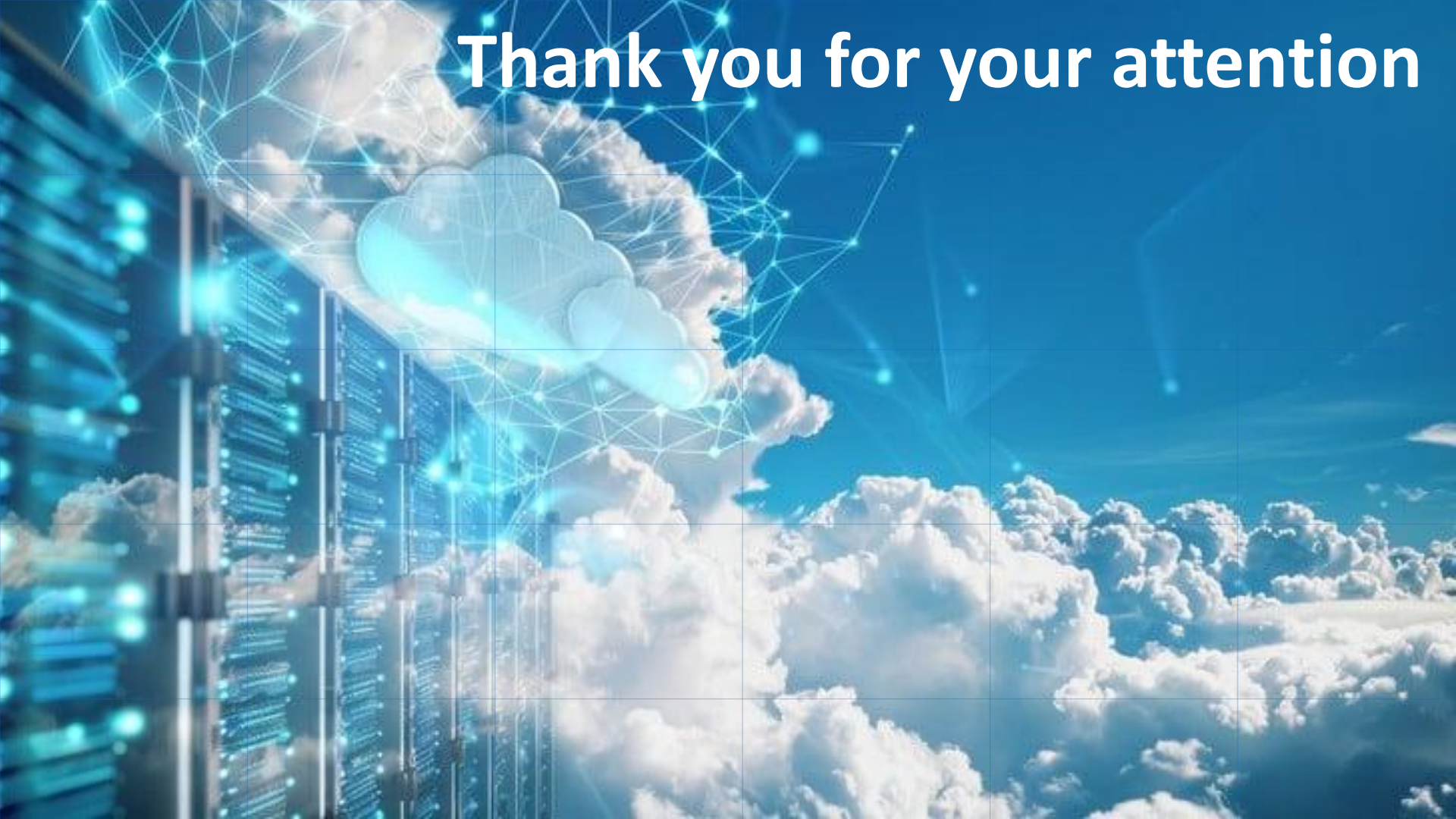
NAVER Gak, South Korea



The citadel Nevada

Cloud Computing IT Infrastructure




Thank you for your attention